Incidence of Drug Price Regulations

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Abstract

Prescription drug prices are determined through a chain of interactions among manufacturers, insurers, and consumers. A regulator seeking to lower drug prices must choose which point in the chain to regulate and which drugs to target. How are the effects of drug price regulations distributed among market participants, and how does that guide a regulator's preferred choices? We develop a model of a drug market with rebate bargaining and community-rated premiums. A net price ceiling transfers surplus from manufacturers to insurers, and a copay cap reaches drug users directly with insurers and manufacturers splitting the cost. Beyond that, consumers may be affected through premiums, which only move when drug users are marginal enrollees. A consumer-welfare-maximizing regulator should cap copays, while a spending-minimizing regulator can use either instrument. Copay caps should target drugs with high out-of-pocket prices, many users, and elastic utilization; net price ceilings should target high net spending. Applied to insulin in Medicare Part D, the model implies that Medicare's $35 insulin copay cap raises fills by 3% at a cost borne by insurers and manufacturers rather than by other consumers.

Keywords: drugs, price regulation, incidence, rebate bargaining, Medicare Part D, insulin
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1 Introduction

High prescription drug spending is one of the most visible sources of financial strain in the United States health care system. According to the Centers for Medicare & Medicaid Services (CMS), consumers and insurers spent a combined total of $406 billion, or $1,200 per American, on retail prescription drugs in 2022 (CMS, 2023b). The prices behind that spending are determined through a chain of interactions among manufacturers, insurers, and consumers. Manufacturers set list prices. Insurers, usually through pharmacy benefit managers, negotiate rebates that determine the net prices they pay. Consumers pay premiums for insurance coverage and out-of-pocket prices for drug purchases.

A regulator seeking to lower drug prices must therefore choose which point in this chain to regulate and which drugs to target. For example, out-of-pocket prices are frequently regulated, especially for insulin: Medicare has capped insulin copays at $35 since 2023 (CMS, 2023a), and 29 states cap insulin copays for state-regulated commercial insurers (ADA, 2026). Prices that insurers pay manufacturers are often regulated as well. The Medicare Drug Price Negotiation Program sets maximum fair prices which, starting in 2026, constrain prices paid by Medicare for a set of high-spending drugs (CMS, 2024). Similarly, Medicaid's most-favored-nation pricing program was launched in 2026 and benchmarks Medicaid prices for participating drugs based on prices in other countries (CMS, 2026). Because each of these regulations indirectly impacts the rest of the chain, equilibrium effects can be difficult to predict, and headline numbers can mislead: the Drug Price Negotiation Program's first maximum fair prices were 62% below list prices, yet CMS estimates they only lowered net spending by 22% because rebates had already captured most of the difference.

In this paper, we ask two related questions. First, how are the effects of drug price regulations distributed among consumers, insurers, and manufacturers? Second, how does that guide a regulator's preferred choices of policy instrument and targeted drug? The answers depend on market structure and demand: the relative bargaining power of insurers and manufacturers, insurers' market power in setting premiums, which consumers are marginal to their plans, how many patients would pay list prices without coverage, and how elastic demand is at the pharmacy and in the cash market.

We study these questions in a model with one manufacturer of a drug, one insurer (which we then extend to multiple competing insurers), and two types of consumers: users of the drug and non-users. The list price of the drug is taken as exogenous, and the manufacturer and insurer bargain over the rebate and resulting net price. The insurer sets one community-rated premium for all consumers, and out-of-pocket prices are determined by an exogenous copay. Consumers choose whether to enroll in insurance, and users choose whether to purchase the

drug. The effect of any policy on a particular market participant can be decomposed into a *direct* channel (changes in prices and quantities with the rebate and premium held fixed), a *premium* channel (equilibrium effects due to premium changes), and a *bargaining* channel (equilibrium effects due to rebate changes). We focus on two policy instruments: a copay cap and a net price ceiling.

A copay cut reaches drug users directly, and its cost is shared by the insurer and manufacturer. We show that the bargained net price equals the manufacturer's marginal cost, plus its cash-market outside option per insured fill, plus a share $1-\eta$ of the joint gains from coverage per insured fill, where η is the insurer's relative bargaining power. Copay revenue is part of those joint gains, so a \$1 copay cut lowers the net price by about $1-\eta$ dollars: bargaining power determines how the firms divide the cost. Non-users pay through premiums only when users are marginal to the plan. That partially depends on the policy's scope: users who are inframarginal to coverage are still marginal between competing plans, so a copay cap adopted by one plan is paid by that plan's non-users while a market-wide mandate is not. With insurer market power, a copay cut can reduce total surplus even when production is costless by raising the premium that non-users face.

A net price ceiling directly transfers surplus from the manufacturer to the insurer. Consumers benefit from this transfer only through a lower premium, which responds in proportion to drug users' share of *marginal* enrollees due to community rating. When users are inframarginal, as insulin users often are, the savings stay with the insurer.

A third possible instrument is a list price ceiling; however, we show that a list price ceiling above the prevailing net price does nothing, while one below it is equivalent to a net price ceiling. A list price ceiling is notable only if enough patients buy in the cash market, where it lowers the cash price and changes the manufacturer's disagreement payoff.

These incidence results yield simple guidelines for regulators. A regulator maximizing consumer welfare should cap copays, which would reach drug users directly, rather than the price insurers pay, which would mainly benefit insurers. A regulator minimizing drug spending can use either instrument: a net price ceiling lowers spending directly, and a copay cap lowers it through bargaining when demand is inelastic. For copay caps, the most effective drugs to target are those with high out-of-pocket prices, a large user base, and elastic utilization. For net price ceilings, the most effective drugs to target are those with high net spending, rather than gross spending. This highlights the shortcoming in the Drug Price Negotiation Program's selection rule, which ranks drugs by gross spending: if a drug's high gross spending is already offset by large rebates, a price ceiling may have no effect.

We apply the model to the empirical setting of insulin in Medicare Part D. Insulin is clinically essential, produced by three dominant manufacturers (Knox, 2020), sold through rebate-

based formulary contracts, and the subject of many drug affordability policies. Using 2022 Part D claims linked to plan formularies, we document the importance of every step along the pricing chain: the mean insulin fill had a list price of $664.50 and an out-of-pocket price of $30.90. We estimate demand for insulin products and calibrate the model to these data, the mean premium of stand-alone plans, the premium pass-through estimates of Cabral, Geruso, and Mahoney (2018), and the utilization response to Medicare's insulin copay cap estimated by Myerson et al. (2026).

Three findings stand out. First, a $35 copay cap binds for 35% of insulin users. The copay cap raises their fills by 9.2% and all users' fills by 3.0%, gives affected users $332 per year, and costs the insurer and the manufacturer $2.50 each per beneficiary-year, while non-users pay nothing. Each dollar of copay removed lowers the bargained net price by about 56 cents, so the manufacturer finances about half of any copay cap through rebates. The manufacturer's share of the cost ranges from 20% to 80% with bargaining power, and non-users' share rises from about 1% to 11% with high premium pass-through and to 44% when users are more marginal enrollees. Second, with competing plans, a copay cap adopted by a single plan raises its insulin-user enrollment by 38% and costs its remaining non-user enrollees $32 per year, 0.17 of users' gain; a market-wide mandate costs non-users $5. Third, a net price ceiling transfers almost all of the manufacturer's loss to the insurer, and consumers receive 0.3 cents per dollar of the transfer, because insulin users are inframarginal enrollees.

In a targeting exercise with archetypal drugs, the drugs a consumer-welfare-maximizing regulator would target and those a spending-minimizing regulator would target differ, and neither follows the Drug Price Negotiation Program's ranking by gross spending. Finally, we validate the model by reproducing three untargeted moments: the realized savings of the Drug Price Negotiation Program's first round; the premium increases and rise in rebates under the Senior Savings Model, which capped insulin copays for participating Medicare plans; and the absence of an average utilization response to the market-wide $35 insulin copay cap alongside a concentrated response among the most exposed consumers.

Related literature. This paper contributes to three literatures. The first studies bargaining in vertically related health care markets: Gowrisankaran, Nevo, and Town (2015) and Ho and Lee (2017) estimate Nash-in-Nash models of hospital–insurer negotiations, Collard-Wexler, Gowrisankaran, and Lee (2019) provide microfoundations, Demirer and Rubens (2025) characterize the welfare effects of buyer and seller power in vertical chains, and Ho and Lee (2026) model contracting between a pharmacy benefit manager and manufacturers over formulary-contingent rebates. We use this framework to compare regulations applied at different points in the chain and to derive a regulator's instrument choice and targeting rules.

The second studies pass-through and insurer market power: state benefit mandates raise premiums (Bailey & Blascak, 2016), Medicare Advantage subsidies pass through incompletely and more so in competitive markets (Cabral et al., 2018), and insurer consolidation raises premiums (Dafny, Duggan, & Ramanarayanan, 2012; Decarolis, Polyakova, & Ryan, 2020). We show that under community rating the pass-through of a drug-specific cost equals the pass-through of a uniform cost scaled by drug users' share of marginal enrollees, an application of the logic of Weyl and Fabinger (2013) that determines whether the pool of consumers pays for targeted benefits.

The third studies prescription drug cost sharing and affordability policy: patients respond to spot prices (Brot-Goldberg, Chandra, Handel, & Kolstad, 2017), cost sharing among the elderly can raise hospital use (Chandra, Gruber, & McKnight, 2010; Hsu et al., 2006), price responsiveness varies widely across Part D drugs (Einav, Finkelstein, & Polyakova, 2018), and insulin copay caps lower out-of-pocket spending with utilization responses concentrated among highly exposed patients (Giannouchos, Ukert, & Buchmueller, 2024; Myerson et al., 2026). Duggan and Scott Morton (2010) show that Part D's expansion of insured demand affected drug prices, and Dafny, Ody, and Schmitt (2017) show that copay coupons raise costs by undermining formulary incentives. We add equilibrium premium and rebate responses to this evidence and show why incidence varies across policies and across market participants.

Outline. Section 2 describes the pricing chain and recent policies and presents motivating facts. Section 3 develops a benchmark model with closed-form incidence. Section 4 presents the full model and characterizes the incidence of each instrument and the role of the list price. Section 5 derives the regulator's instrument choice and targeting rules. Section 6 applies the model to insulin in Medicare Part D, and Section 7 concludes.

2 Institutional Background and Motivating Facts

2.1 Drug Pricing Chain

Three prices matter along the drug pricing chain, which Appendix Figure A1 summarizes. The *list price*, formally the wholesale acquisition cost, is set by the manufacturer. The *net price* is what the manufacturer retains after the confidential rebates that insurers, typically through pharmacy benefit managers, negotiate in exchange for formulary placement (Ho & Lee, 2026). The *out-of-pocket price* is what the insured patient pays at the pharmacy; it is set by the insurer's benefit design, as a fixed copay or as coinsurance on each formulary tier. Patients' cost sharing does not depend on the list price if they face a copay, but may be a share of pre-rebate price if they face a coinsurance. An uninsured patient, or an insured patient whose plan excludes the

drug, faces the list price or a pharmacy price derived from it.

2.2 Drug Price Regulation Policies

Several types of policies have been adopted to improve the affordability of prescription drugs, including insulin. The policies mainly fall into two categories: ones targeting the net prices insurers pay manufacturers and ones targeting downstream out-of-pocket payments. Appendix Table A1 maps each policy to the price it regulates and its corresponding model object.

2.2.1 Net Price Regulations

Most-favored-nation pricing. The Trump administration adopted most-favored-nation (MFN) drug pricing, under which the U.S. should pay similar prescription drug prices as other comparable nations (White House, 2025). By early 2026, most major manufacturers had entered voluntary MFN agreements with the administration (Cubanski, 2026). The agreements deliver MFN prices through three channels: rebates to Medicaid under a payment model launched in 2026 (CMS, 2026), a direct-to-consumer cash channel, and launch prices for new drugs. In the Medicaid channel, the agreements regulate the net price Medicaid pays for participating drugs.

Medicare drug price negotiation. The Inflation Reduction Act (IRA) of 2022 granted the Secretary of Health and Human Services the authority to negotiate prices with drug companies for certain high-expenditure, single-source drugs without generic or biosimilar competition covered under Medicare (Cubanski & Neuman, 2026). The program requires CMS to negotiate maximum fair prices for the set of selected drugs, taking into account the drug's clinical benefits and manufacturer-specific data elements. The maximum fair price cannot exceed the lesser of the drug's prior enrollment-weighted net price and a percentage of its average manufacturer price, so it is benchmarked to the net price rather than the list price. If negotiations fail, manufacturers may be subject to an excise tax or can choose to withdraw their drugs from coverage under Medicare and Medicaid to avoid the excise tax. The first ten negotiated prices took effect in January 2026 (CMS, 2024); drugs selected for 2028 include drugs to treat type 2 diabetes, HIV, and prostate cancer.

2.2.2 Out-of-Pocket Price Regulations

State copay caps. Twenty-nine states and the District of Columbia have enacted legislation capping patient out-of-pocket costs for insulin in state-regulated commercial plans (ADA, 2026). Colorado was among the first, enacting a $100 monthly copay cap for insulin that took effect in 2020. Giannouchos et al. (2024) find that Colorado's copay cap increased medication

adherence and reduced diabetes-related complications, providing early evidence that out-of-pocket cost reductions can meaningfully improve patient outcomes.

Medicare Senior Savings Model. Beginning January 1, 2021, CMS tested the impact of allowing Medicare Prescription Drug Plans (PDPs) and Medicare Advantage Prescription Drug plans (MA-PDs) to voluntarily cap patient insulin cost sharing at $35 per month's supply as a supplemental benefit (CMS, 2025; Cubanski & Neuman, 2024). Participation was voluntary for plans, and around half of all Part D plans joined. The copay cap did not apply to all products: participating plans only had to cover one of each dosage form and insulin type. The model ran through December 2022 and was superseded by the IRA's mandatory provisions.

Inflation Reduction Act insulin copay cap. Beginning in January 2023, the IRA mandated that all Medicare Part D plans cap monthly cost sharing for covered insulin products at $35, with no deductible applying (CMS, 2023a; Cubanski & Neuman, 2024). Beginning in July 2023, the copay cap also applied to Part B. Prior to the copay cap, the average annual out-of-pocket spending on insulin among Medicare Part D enrollees was $572 in 2020 (Cubanski & Damico, 2022); 25% of Part D enrollees spent more than $800, and 5% spent more than $1,700. The IRA's $35 copay cap is therefore binding for a meaningful share of insulin users. From 2026, the copay cap for a covered insulin product is the lesser of $35, 25% of its maximum fair price, and 25% of the plan's negotiated price (CMS, 2025).

2.3 Three Motivating Facts

Fact 1: List-price discounts overstate net price reductions. Table 1 reports the ten drugs in the first negotiation cycle. Negotiated prices were 38% to 79% below 2023 list prices, and 62% below on a spending-weighted basis. Applied to the drugs' $56.2 billion of 2023 gross Part D spending, these discounts would imply a $34.7 billion reduction. CMS estimates instead that the negotiated prices would have reduced 2023 net spending by $6 billion, or 22%. The implied baseline net spending is $27.3 billion, so rebates already amounted to about 51% of gross spending, and only 17% of the headline reduction is a reduction in net spending. This is the pattern the model of Section 4 predicts: what matters is the regulated price relative to the prior *net* price, not relative to the list price.

Fact 2: The benefits of a copay cap are concentrated where it binds. Using Part D data on 2.86 million insulin users, Myerson et al. (2026) find that the $35 copay cap reduced average out-of-pocket cost per 30-day supply from $22.95 to $18.16 (21%) and within-year variation in out-of-pocket cost by 48%, with no increase in average fills, adherence, or persistence among

Table 1: Medicare's First Ten Negotiated Prices: List Discounts versus Net Savings

	Price, 30-day supply ($)		Discount	Gross cost	Users	Gross cost ×
Drug	List (2023)	Negotiated (2026)	from list (%)	2023 ($bn)	(000s)	discount ($bn)
Eliquis	521	231	56	18.3	3,928	10.2
Jardiance	573	197	66	8.8	1,883	5.8
Xarelto	517	197	62	6.3	1,324	3.9
Farxiga	556	178	68	4.3	994	2.9
Januvia	527	113	79	4.1	843	3.2
Entresto	628	295	53	3.4	664	1.8
Stelara	13,836	4,695	66	3.0	23	2.0
Enbrel	7,106	2,355	67	3.0	48	2.0
NovoLog/Fiasp	495	119	76	2.6	785	2.0
Imbruvica	14,934	9,319	38	2.4	17	0.9
All ten			62[a]	56.2	10,509	34.7
CMS estimate of reduction in 2023 net spending: $6.0 billion (22%)						

Notes: Data from CMS (2024). List prices are 2023 wholesale acquisition costs. The last column multiplies each drug's gross cost by its list discount; it is the reduction in spending implied if negotiated prices were compared with list prices. [a] Weighted by 2023 gross cost.

the full cohort. The effects were concentrated among patients with high baseline costs for insulin: in the top decile of baseline out-of-pocket cost (at least $58 per supply), the copay cap lowered out-of-pocket cost by $46.27 (51%) and raised annual fills by 0.8 (8%). The implied arc elasticity of insulin use for highly exposed patients is about 0.16. This is an order of magnitude below product-level elasticities, which include switching among products.

Fact 3: A copay cap shifted costs to manufacturers and to non-users' premiums. Appendix Table A2 reports difference-in-differences estimates from CMS's evaluation of the Senior Savings Model. Insulin users' combined premium and out-of-pocket spending fell by $199–$319 per year in 2021–2022. In contrast, manufacturer rebates and coverage-gap discounts rose in both plan types. Non-users' spending rose by $17–$24 per year in Medicare Advantage drug plans and by $30–$65 in stand-alone plans (PDPs). In PDPs, non-users' out-of-pocket drug spending *fell*, so the entire increase came through premiums, underscoring the importance of considering insurers' premium adjustments in copay cap policies. Participating PDPs also saw insulin-user enrollment rise by 75%–92%, showing that marginal enrollees include insulin users, the condition under which the model predicts that non-users bear the cost of copay caps.

3 Benchmark Model

We begin with a model simple enough to deliver closed-form incidence. It isolates two dimensions of market structure, the insurer's bargaining power against the manufacturer and its market power over consumers, and it shows that the list price matters only through what patients would pay outside coverage.

3.1 Setup

There is one drug produced by a manufacturer (at zero marginal cost), and one insurer that enrolls a population of potential drug consumers of measure 1. The drug's list price $p > 0$ and copay $c \in [0, p]$ are taken as exogenous. For consumers without drug coverage, the list price is also the cash price. Consumers get i.i.d. health shocks and choose whether to buy the drug after the shock realizes.

Let $D(q)$ be each consumer's ex-ante probability of buying the drug, and $W(q)$ be each consumer's ex-ante expected value (in monetary units) from the drug, as functions of the out-of-pocket price q, which varies by insurance coverage. They are both decreasing in the out-of-pocket price, with $W'(q) = -D(q)$. If the drug is covered, the out-of-pocket price is c paid to the insurer, and the insurer pays the manufacturer the net price k per fill. If not, the out-of-pocket price is p paid directly to the manufacturer.

The manufacturer and insurer bargain over the net price k, equivalently the rebate $p - k$. If they disagree, then the drug is not covered. If they agree, then the drug is covered and the insurer purchases the drug for consumers at the net price. In addition, each consumer pays a premium to the insurer. The premium allows the insurer to extract a fraction α of the consumer's value of drug coverage and pass on a fraction $1 - \alpha$ of its costs. Effectively, the premium distributes welfare between the insurer and consumers, such that the insurer receives a fraction α of total insurer and consumer welfare generated by drug coverage, where $\alpha \in [0, 1]$ reflects the insurer's market power.

Define the coverage surplus net of the insurer's drug cost,

$$S(k) \equiv \underbrace{W(c) - W(p)}_{\text{Value of coverage}} - \underbrace{(k - c)D(c)}_{\text{Insurer drug costs}} . \tag{1}$$

Appendix Table A3 summarizes manufacturer, insurer, and consumer welfare depending on whether the manufacturer and insurer agree or disagree: the insurer's welfare is $\alpha S(k)$ and consumers' is $W(p) + (1 - \alpha)S(k)$.

Define $k^{\max}$ as the highest value of k such that the insurer is willing to agree, and $k^{\min}$ as the lowest value of k such that the manufacturer is willing to agree. They satisfy

$$k^{\max} = c + [W(c) - W(p)]/D(c) \tag{2}$$

$$k^{\min} = pD(p)/D(c) \tag{3}$$

Assume $k^{\min} \leq k^{\max}$ (i.e., there exists a value of k such that the manufacturer and insurer are both willing to agree). Consider two cases of bargaining, corresponding to different levels of the insurer's relative bargaining power:

- Case 1: The manufacturer makes a take-it-or-leave-it offer to the insurer. Thus, it offers $k = k^{\max}$, which is accepted.
- Case 2: The insurer makes a take-it-or-leave-it offer to the manufacturer. Thus, it offers $k = k^{\min}$, which is accepted.

3.2 Incidence

We consider the incidence of a net price ceiling and of the copay under the two bargaining cases. The proof in Appendix A also derives the incidence of a list-price change, which shows how the net price readjusts to upstream and downstream changes.

Proposition 1 (Benchmark incidence)**.** *Total surplus under coverage is $TS = W(c) + cD(c)$; it does not depend on p or k, and $dTS/dc = cD'(c) \leq 0$. Let k^* denote the bargained net price ($k^* = k^{\max}$ in Case 1, $k^* = k^{\min}$ in Case 2), and consider a net price ceiling $\bar{k} \geq k^{\min}$. The incidence of the ceiling and of the copay is:*

	Case 1: Manufacturer makes offer	Case 2: Insurer makes offer
∂ Manufacturer$/\partial\bar{k}$	$D(c)$	0
∂ Insurer$/\partial\bar{k}$	$-\alpha D(c)$	0
∂ Consumers$/\partial\bar{k}$	$-(1-\alpha) D(c)$	0
∂ Manufacturer$/\partial c$	$cD'(c)$	0
∂ Insurer$/\partial c$	0	$\alpha cD'(c)$
∂ Consumers$/\partial c$	0	$(1-\alpha) cD'(c)$

In Case 1 the ceiling binds for $\bar{k} < k^{\max}$; in Case 2, $k^* = k^{\min}$, so the ceiling does not bind. If $\bar{k} < k^{\min}$, the manufacturer withholds the drug from coverage and total surplus falls by $G \equiv [W(c) + cD(c)] - [W(p) + pD(p)]$.

Two features of Proposition 1 reappear in the full model.

First, the net price ceiling is a pure transfer: it does not change the total surplus since it does not affect drug demand, and it only reallocates the payment $kD(c)$ on covered units.[1] The transfer occurs only under Case 1, where the ceiling moves $D(c)$ per unit of $\bar{k}$ from the manufacturer to the insurer, who passes a share $1-\alpha$ to consumers through premiums. Under insurer power (Case 2), the net price already sits at the manufacturer's outside option, so the ceiling does not bind. If it is set below $k^{\min}$, the manufacturer withholds the drug and total surplus falls by G.

Second, the copay is the only price with efficiency consequences since it directly affects drug utilization, and its deadweight loss $cD'(c)$ falls on whichever party is not held to its outside option in the bargain.

The benchmark leaves out what makes drug regulation contentious in practice: insurance enrollment responds to premiums, premiums are community-rated across users and non-users of a drug, and bargaining outcomes are interior. We add these features next.

4 Model of a Drug Market

4.1 Environment

We use a model of rebate bargaining, insurance demand, and drug demand, similar to Ho and Lee (2026), to illustrate how policies targeting different points in the drug pricing chain affect prices, enrollment, and welfare. Insurance enrollment and drug purchase probabilities are endogenous: the premium, which depends on the negotiated rebate, influences insurance enrollment decisions, and drug purchase probabilities vary depending on an individual user's insurance status.

Consumers. A unit mass of beneficiaries consists of a share δ of potential drug users ("users") and a share $1-\delta$ of non-users. All beneficiaries choose whether to enroll in insurance, while users additionally decide whether to purchase the drug. Each enrolled user fills a prescription with probability $D(c)$ at copay c and receives expected value $W(c)$ from access to the drug, with $W' = -D$.

Consumers choose whether to purchase the drug based on their insurance status: outside insurance, users can buy the drug at the list price p_L with purchase probability $D_0(p_L)$ and value $W_0(p_L)$, $W_0' = -D_0$. We call this the cash market. Its size, $\delta(1-s_D)D_0(p_L)$, is endogenous: enrollment s_D responds to the premium and to $W_0(p_L)$, so a lower list price both raises

[1] The effects will be different for drugs covered under coinsurance, which causes patient out-of-pocket costs to depend on the list price.

purchases per uninsured user and makes going uninsured more attractive. If bargaining fails the drug is not covered and every user faces the cash price. The insurer charges a community-rated premium P. Consumers consider the expected value of drug purchases when choosing insurance, and non-users may still derive value from insurance: all beneficiaries value the plan's other coverage (e.g., from covering other drugs) at g. With logit enrollment shocks of scale $1/\lambda$, enrollment probabilities are

$$s_D = \Lambda\big(\lambda[g + W(c) - W_0(p_L) - P]\big), \qquad s_N = \Lambda\big(\lambda[g - P]\big), \tag{4}$$

where Λ is the logistic function. Consumer surplus per user and per non-user is

$$CS_D = \tfrac{1}{\lambda}\log\big(e^{\lambda(g+W(c)-P)} + e^{\lambda W_0(p_L)}\big), \qquad CS_N = \tfrac{1}{\lambda}\log\big(e^{\lambda(g-P)} + 1\big). \tag{5}$$

Firms. Let k denote the net price paid by the insurer to the manufacturer per fill. Equivalently the rebate rate is $r = 1 - k/p_L$, so $k \in [0, p_L]$. Profits are

$$\Pi_I = \underbrace{\delta s_D\big[P - (k-c)D(c)\big]}_{\text{Users}} + \underbrace{(1-\delta)s_N P}_{\text{Non-users}}, \tag{6}$$

$$\Pi_M = \delta\big[\underbrace{s_D(k-m)D(c)}_{\text{Insured users}} + \underbrace{(1-s_D)(p_L-m)D_0(p_L)}_{\text{Uninsured users}}\big], \tag{7}$$

where $m \geq 0$ is the marginal cost of production and dispensing. Coverage is binary, either covered or not covered; we model negotiation over insulin only and hold the negotiation of non-insulin drugs fixed.

Timing. The model proceeds in four steps:

1. The manufacturer and insurer bargain over k.
2. Under community rating, the insurer sets the premium P to maximize profit.
3. Beneficiaries choose insurance coverage.
4. Users choose whether to fill their prescription.

Instruments and channels. The regulator's instruments are the copay c and the net price ceiling $\bar{k}$. The list price p_L is an exogenous parameter of the environment, and Section 4.5 shows when it matters. Let V denote an outcome of interest (manufacturer profit, insurer

profit, and consumer welfare) and x denote a policy instrument (copay cap, net price ceiling). Writing k^* for the equilibrium negotiated net price and $P^*(k, x)$ for the premium given k,

$$\frac{dV}{dx} = \underbrace{\frac{\partial V}{\partial x}}_{\text{direct}} + \underbrace{\frac{\partial V}{\partial P}\frac{\partial P^*}{\partial x}}_{\text{premium}} + \underbrace{\Big[\frac{\partial V}{\partial k} + \frac{\partial V}{\partial P}\frac{\partial P^*}{\partial k}\Big]\frac{dk^*}{dx}}_{\text{bargaining}}. \tag{8}$$

The equation shows that a policy instrument induces three types of changes to welfare.

1. The *direct effect*, where the change in the copay or the net price ceiling directly impacts welfare or profit, holding premium and rebate constant.

2. The *premium channel*, where the policy lever affects the premium set by the insurer. The term is governed by the insurer's pass-through rate $\partial P^*/\partial x$, which describes how much of a change in the insurer's net drug cost is passed to premiums. Under community rating, changes in premiums reflect both changes in costs and changes in the composition of enrollees. When there are multiple insurers, the pass-through rate further captures insurer market power, and with more competition pass-through is closer to full.

3. The *bargaining channel*, where the policy lever impacts the negotiated rebate by changing the insurer's and manufacturer's bargaining positions. This channel is governed by the bargaining sensitivity dk^*/dx, which captures how much lever x shifts the negotiated net price and depends on how x moves each side's gains from agreement and on the bargaining weight η, and by the pass-through of the renegotiated rebate into the premium, $\partial P^*/\partial k$.

In the quantitative analysis we compute the discrete analog: we first change the policy holding the net price and premium fixed, then re-optimize the premium, then renegotiate the net price.

4.2 The Bargained Net Price

In the first stage, the manufacturer and insurer bargain over the net price.

If the manufacturer and insurer reach agreement at net price k, the insurer sets premium $P(k)$, with enrollment shares $s_D(k) \equiv s_D\big(P(k)\big)$ and $s_N(k) \equiv s_N\big(P(k)\big)$ of drug users and non-users. Under disagreement, the insurer does not cover the drug and sets premium P^0 to maximizes profit without drug expenditures, with enrollment shares $s_D^0 \equiv s_D(P^0)$ and $s_N^0 \equiv s_N(P^0)$ and profit $\Pi_I^0 = P^0\big[\delta s_D^0 + (1-\delta)s_N^0\big]$.

The manufacturer's disagreement payoff is $\Pi_M^0 = \delta(p_L - m)D_0(p_L)$, its profit when users can access the drug only at its list price. The insurer's disagreement payoff is $\Pi_I^0 - \varphi$, where $\varphi \geq 0$ captures any further loss from not covering the drug, such as a formulary requirement

or the loss of users' other business, that a one-drug model otherwise omits. We set $\varphi = 0$ in the baseline case; Section 6 calibrates it to external evidence on rebates.

The net price solves

$$k^* = \underset{k\in[0,\bar{k}]}{\operatorname{argmax}}\ (1-\eta)\log\left[\Pi_M(k) - \Pi_M^0\right] + \eta\log\left[\Pi_I(k) - \Pi_I^0 + \varphi\right], \tag{9}$$

where η is the insurer's bargaining weight and $\bar{k} \le p_L$ is any net price ceiling ($\bar{k} = p_L$ if there is none). The joint gains from agreement are

$$\begin{aligned} J(k) &\equiv \left[\Pi_I(k) - \Pi_I^0 + \varphi\right] + \left[\Pi_M(k) - \Pi_M^0\right] \\ &= \delta s_D(k)\big[P(k) + (c-m)D(c) - (p_L - m)D_0(p_L)\big] + (1-\delta)s_N(k)\,P(k) \\ &\quad - P^0\big[\delta s_D^0 + (1-\delta)s_N^0\big] + \varphi. \end{aligned} \tag{10}$$

The negotiated net price k is a transfer between the insurer and the manufacturer and does not appear explicitly in J. However, it affects J through the insurer's premium response. Using the insurer's first-order condition for P,

$$J'(k) = \delta\,\frac{\partial s_D}{\partial P}\,P'(k)\left[(k-m)D(c) - (p_L - m)D_0(p_L)\right], \tag{11}$$

so J is independent of k only if drug users' enrollment does not respond to the premium or the manufacturer earns the same margin per drug user under agreement and disagreement. Total surplus is $TS = \delta CS_D + (1-\delta)CS_N + \Pi_I + \Pi_M$.

Lemma 1 provides some intuition on how the negotiated net price is determined.

Lemma 1 (Net price as a surplus share)**.** *Suppose the Nash objective is single-peaked in k and the premium does not respond to k at the optimum ($\partial P^*/\partial k = 0$). Let $Q \equiv \delta s_D D(c)$ denote fills from insured users. Then the unconstrained bargained net price is*

$$k^u = m + \underbrace{(p_L - m)\frac{D_0(p_L)}{D(c)}}_{\text{outside option per fill}} + \underbrace{(1-\eta)\frac{J}{Q}}_{\text{share of gains per fill}}, \tag{12}$$

and $k^* = \min\{k^u, \bar{k}\}$. When $\partial P^*/\partial k \neq 0$, the last term is multiplied by a factor that converges to one as users' share of marginal enrollees goes to zero.

Equation (12) has a simple interpretation. The manufacturer is paid its marginal cost, plus what it would earn per insured fill by selling in the cash market instead, plus its bargaining share of the joint gains from coverage per insured fill. Copay revenue $cD(c)$ enters the joint gains: it is paid by patients into the vertical chain and is shared between the parties.

This yields a useful approximation. When users are inframarginal and $D_0 = 0$, $J/Q = c - m + \bar{\pi}$, where $\bar{\pi}$ collects premium gains per fill that do not depend on c to first order. Hence $dk^u/dc \approx 1 - \eta$: each dollar of copay removed lowers the net price by about $1 - \eta$ dollars.

More generally, if we assume that premiums can adjust in response to the net price, then mapping to the setting of Horn and Wolinsky (1988), the insurer's downstream market power affects upstream bargaining outcomes through a premium pass-through: if pass-through is high, the large enrollment and utilization responses make concessions less costly for the manufacturer, while if pass-through is low, a larger rebate is more costly for it.

4.3 Regulating the Copay

For an insurer, a copay cap represents a change in cost per enrolled user. The insurer can pass the cost downstream through a single premium under community rating, or upstream through the negotiated net price. We first characterize the downstream premium pass-through for any change in users' cost, then show how a copay cap changes that cost once the net price adjusts, and finally derive its incidence on users and non-users.

The intuition behind the downstream pass-through has two steps. First, if we consider a cost τ that the insurer incurs equally for every enrollee (e.g., administrative fees), its pass-through to the premium, $\rho \equiv \partial P^*/\partial \tau$, reflects how much the insurer can raise the premium without losing too many enrollees. Second, a cost that falls only on users matters for the premium only if users are among the marginal enrollees that the insurer is competing for. When setting the premium, the insurer weighs each group's cost by that group's share of the enrollees who would leave after a small premium increase, which we denote ω_D. Users who would enroll at almost any premium carry little weight, even if they make up a large share of the plan.

Proposition 2 (Downstream premium pass-through). *Let $u_D = (k-c)D(c)$ denote the insurer's drug cost per enrolled user, and $s'_g = \partial s_g/\partial P$ denote the response of group g's enrollment to the premium. Holding fixed k and the enrollment functions, the premium's response to a change in u_D is*

$$\frac{\partial P^*}{\partial u_D} = \rho\, \omega_D, \tag{13}$$

where $\rho = \partial P^*/\partial \tau$ is the pass-through of a cost τ incurred for every enrollee, and

$$\omega_D \equiv \frac{\delta s'_D}{\delta s'_D + (1-\delta)s'_N} = \frac{\delta s_D(1-s_D)}{\delta s_D(1-s_D) + (1-\delta)s_N(1-s_N)} \tag{14}$$

is users' share of marginal enrollees. The second equality holds under logit enrollment with a common premium coefficient. Users' share of marginal enrollees can be far below their population share δ: $\omega_D \to 0$ as $s_D \to 1$.

For example, if users make up 10% of the population ($\delta = 0.1$) and enroll with probability $s_D = 0.95$, while non-users enroll with probability $s_N = 0.6$, users make up only about 2% of marginal enrollees.

A copay cap raises u_D by $D(c) + (k - c)|D'(c)|$ per dollar at a given net price: the insurer pays the forgone copay on existing fills and its net cost of fills induced by the copay change. Part of this cost is passed upstream. Because the copay enters the joint gains from agreement, the bargained net price falls by $\kappa \equiv dk^*/dc$ per dollar of copay reduction (Lemma 1), where $\kappa \approx 1 - \eta$ when users are inframarginal and $\kappa = 0$ when the net price ceiling binds. The insurer's cost per user therefore rises by $(1 - \kappa)D(c) + (k - c)|D'(c)|$, and by Proposition 2 the premium rises by[2]

$$\frac{dP^*}{d(-c)} = \rho\, \omega_D\big[(1 - \kappa)D(c) + (k - c)|D'(c)|\big]. \tag{15}$$

This premium increase falls on all enrollees, while only users benefit from the lower copay. By the envelope theorem, $\partial CS_D/\partial P = -s_D$, $\partial CS_N/\partial P = -s_N$ and $\partial CS_D/\partial c = -s_D D(c)$. Hence,

$$\frac{dCS_N}{d(-c)} = -s_N \frac{dP^*}{d(-c)}, \qquad \frac{d[\delta CS_D + (1 - \delta)CS_N]}{d(-c)} = \delta s_D D(c) - \big[\delta s_D + (1 - \delta)s_N\big]\frac{dP^*}{d(-c)}. \tag{16}$$

Non-users lose only through the premium, while users lose through the premium but gain $D(c)$ per enrollee directly. Substituting (15) into (16) gives the following corollary:

Corollary 1 (Users gain, pool loses)**.** *Ignoring the valuation shift, a copay cut raises users' surplus if and only if*

$$D(c) > \rho\, \omega_D\big[(1 - \kappa)D(c) + (k - c)|D'(c)|\big]. \tag{17}$$

A copay cut lowers aggregate consumer surplus if and only if

$$\delta s_D D(c) < \big[\delta s_D + (1 - \delta)s_N\big]\rho\, \omega_D \big[(1 - \kappa)D(c) + (k - c)|D'(c)|\big]. \tag{18}$$

If users and non-users are equally marginal ($s_D = s_N$, so $\omega_D = \delta$) and induced fills are negligible, the second condition requires $\rho(1 - \kappa) > 1$, which fails whenever enrollment demand is log-concave, as under the logit. Aggregate consumer losses therefore require users to be disproportionately marginal enrollees or induced use to be costly to the insurer.

Proposition 2 and Corollary 1 connect the model to the evidence. In a single community-rated pool where users have large coverage value and are inframarginal, the cost of a copay cut is borne mostly by the insurer and, through bargaining, by the manufacturer, rather than through premium pass-throughs. In comparison, when plans compete and users sort toward

[2] A copay cap also raises users' valuation of coverage. We ignore this channel; under the logit, its premium effect also vanishes as $s_D \to 1$.

generous plans, users become marginal to each plan and the premium rises for everyone in the plan. The Senior Savings Model is a case of competing plans: insulin-user enrollment in participating PDPs rose by around 75%, and non-users' premiums rose (Appendix Table A2). Section 6.9 extends the application to competing plans and shows that the same proposition, applied plan by plan, reproduces this pattern for a plan-level copay cap but not for a market-wide mandate.

4.4 Regulating the Net Price

A net price ceiling $\bar{k}$, a cap on the price the insurer pays after rebates such as Medicare's maximum fair price or a state upper payment limit, constrains the bargain directly.

Proposition 3 (Net price ceiling)**.** *Suppose the Nash objective is single-peaked in k. Then:*

(i) $k^* = \min\{k^u, \bar{k}\}$. *The ceiling binds if and only if* $\bar{k} < k^u$, *and when it binds the parties settle at the ceiling, with no further rebate negotiated below it.*

(ii) Define $S = \delta s_D + (1-\delta)s_N$ *as total enrollment. When the ceiling binds, a \$1 reduction in* $\bar{k}$ *raises insurer profit by Q, lowers the premium by* $\rho\,\omega_D D(c)$, *raises aggregate consumer surplus by* $S\rho\,\omega_D D(c)$, *and lowers manufacturer profit by Q less a term of order* ω_D^2 *from the enrollment the lower premium induces.*

(iii) If users are inframarginal ($\omega_D = 0$*), the premium does not respond and the ceiling is a pure transfer of Q per dollar from the manufacturer to the insurer; consumers and total surplus are unchanged. If users are marginal (*$\omega_D > 0$*), the premium falls for every enrollee, so users and non-users both gain, and total surplus rises.*

Part (i) describes how a binding maximum fair price interacts with rebates: with $\bar{k} < k^u$ the Nash objective is increasing in k below the ceiling, so bargaining stops at $\bar{k}$ and no rebate survives, which is what industry analyses expect (Milliman, 2024). Because the maximum fair price is benchmarked to the enrollment-weighted *average* net price and plans must cover selected drugs, plans that had negotiated deeper-than-average rebates lose the exclusion threat behind them and their net prices rise to $\bar{k}$; rebates can survive below the maximum fair price only where selected drugs compete therapeutically, a margin absent from our one-drug model.

Parts (ii) and (iii) describe incidence. A \$1 reduction in the ceiling saves the insurer \$1 on each of its Q insured fills. The insurer also lowers its premium, but because the premium was already profit-maximizing, this adjustment does not generate a first-order change to its profit. The insurer therefore keeps the full saving Q regardless of who is marginal. Whether consumers also gain depends on ω_D. When users are inframarginal, the premium does not move, and the ceiling only transfers surplus from the manufacturer to the insurer. When users

are marginal, the premium falls for every enrollee. This mirrors the copay cap, which raises non-users' premiums when users are marginal.

4.5 The Role of the List Price

The list price is not an instrument in the model, but policies and public discussion are often framed in terms of it. The next result states when it matters.

Proposition 4 (List-price neutrality)**.** *Suppose there is no cash market ($D_0 \equiv 0$, $W_0 \equiv 0$), cost sharing is a fixed copay, and the Nash objective is single-peaked in k. Then:*

(i) The unconstrained bargained net price $k^u(c)$ does not depend on p_L.

(ii) For $p_L \geq k^u$, every equilibrium outcome (premium, enrollment, utilization, all profits and surpluses) is independent of p_L. The rebate rate $r^* = 1 - k^u/p_L$ absorbs any change in the list price.

(iii) A reduction in the list price to $\bar{p} < k^u(c)$, whether by the manufacturer or by regulation, lowers the net price to $\bar{p}$ and has the same effects as a net price ceiling $\bar{k} = \bar{p}$. A reduction that leaves $p_L \geq k^u(c)$ has no effect.

Proposition 4 formalizes why list-price discounts are not savings. The relevant statistic is the gap between the regulated price and the pre-policy *net* price, not the list discount. It also says when a lower list price substitutes for a net price ceiling: only once the list price falls below k^u, and only if no payoff depends on the list price. Neutrality fails whenever some revenue or some patient price is tied to the list price: through cash or uninsured purchases, through coinsurance set as a share of a list-based price, or through statutory formulas indexed to it. Our model has a fixed copay throughout, so coinsurance lies outside it; the cash market is in the model, and Section 6 quantifies it. The next result characterizes the cash-market case.

Proposition 5 (List prices and the outside option)**.** *Suppose there is a cash market and the insurer has all the bargaining power ($\eta = 1$). Then*

$$k^* = m + (p_L - m)\frac{D_0(p_L)}{D(c)}, \tag{19}$$

$$\frac{dk^*}{dp_L} = \frac{D_0(p_L)}{D(c)}\Big[1 - \epsilon_0(p_L)\frac{p_L - m}{p_L}\Big], \qquad \frac{dk^*}{dc} = -(p_L - m)\frac{D_0(p_L)D'(c)}{D(c)^2} \geq 0, \tag{20}$$

where $\epsilon_0 = -p_L D_0'(p_L)/D_0(p_L)$. If cash demand is elastic ($\epsilon_0 > p_L/(p_L - m)$), a lower list price raises the net price. A net price ceiling, which leaves the cash price unchanged, has no such effect.

The deviation from neutrality is governed by $D_0/D(c)$, cash purchases relative to insured fills under disagreement, not by the share of users uninsured in equilibrium, because under disagreement every user faces the cash price. A cash market that looks small can therefore imply a large outside option when the list price is high: in the calibration of Section 6, a 2% purchase probability at a \$664.50 list price gives $D_0/D(c) = 3.6\%$ and a disagreement payoff equal to 52% of the manufacturer's agreement profit. A net price ceiling leaves the list price, and hence the outside option, unchanged; a lower list price changes the outside option as well, which is why it can move the net price in either direction.

4.6 Efficiency with Insurer Market Power

In the benchmark model, a lower copay always raised total surplus when production is costless. With insurer market power over enrollment, this need not hold. The intuition is that with insurer market power, the premium is priced above cost, so too few people enroll. Thus, if a lower copay raises the premium, the enrollees it pushes out of the plan reduce total surplus, which can offset the gains from higher utilization.

Proposition 6 (Total surplus and the copay). *Let $\Omega_D \equiv P+(c-m)D(c)-(p_L-m)D_0(p_L)$ denote the payment to firms from an additional user enrollee that the enrollee does not internalize. Then*

$$\frac{dTS}{dc} = \underbrace{\delta s_D(c-m)D'(c)}_{\text{utilization}} + \underbrace{\delta\frac{\partial s_D}{\partial c}\Omega_D}_{\text{user enrollment}} + \underbrace{\left[\delta s_D'\,\Omega_D + (1-\delta)s_N' P\right]\frac{dP^*}{dc}}_{\text{premium distortion}}, \tag{21}$$

where dP^*/dc is the total premium response, including the effect of renegotiation. The net price affects total surplus only through the premium.

The first two terms are the familiar gains from lower cost sharing: more valuable use and more user enrollment. They are positive for a copay reduction when $c > m$ and $\Omega_D > 0$. The third term is a second-best effect. The premium exceeds the social cost of covering a non-user (zero here), so enrollment is already too low. A copay cut that raises the premium ($dP^*/dc < 0$) pushes further marginal non-users out, reducing surplus. The total-surplus-maximizing copay can therefore exceed marginal cost even without moral hazard, whenever pass-through to marginal enrollees is large. Because the net price enters total surplus only through the premium, a net price ceiling changes total surplus only by the enrollment it induces, which is why Proposition 3(iii) calls it a pure transfer when users are inframarginal.

4.7 What Changes Relative to the Benchmark

In the benchmark of Section 3, a net price ceiling is the only instrument that transfers the manufacturer's rent to consumers, which occurs under manufacturer bargaining power. Con-

sumers receive a share $1-\alpha$ of every dollar the ceiling removes, while a copay cut leaves consumers no better off. The full model reverses this.

Comparing instruments requires a common yardstick, since a dollar off an upstream price and a dollar off a copay are not comparable policies. We use the consumer surplus delivered per dollar of rent extracted from the manufacturer,

$$R \equiv \frac{\Delta CS_A}{-\Delta \Pi_M}, \tag{22}$$

the relevant trade-off when manufacturer profit has a shadow value through innovation or participation. A net price ceiling delivers $R = S\rho\,\omega_D/(\delta s_D)$ and a copay cap delivers $R = 1/\big[\kappa - (k-m)|D'(c)|/D(c)\big]$, with $\kappa = dk^*/dc$. Appendix Table A4 evaluates both at maximum stringency. In the benchmark, consumers capture essentially all of the manufacturer's rent through the upstream price ($R = 1.00$ and 1.25 in the two protocols) and almost nothing through the copay. In the calibrated model the ranking inverts: driving the price insurers pay to zero removes \$11.38 per beneficiary-year from the manufacturer, of which consumers receive \$0.04, while eliminating the copay delivers \$10.79 to consumers at a cost of \$5.15 to the manufacturer, so $R = 2.10$.

Two features of the benchmark account for the difference. Its upstream price works through a large cash margin, which our baseline sets to zero, and its premium mechanically returns a share of any surplus to consumers, whereas in the full model the premium responds to a drug-specific cost only in proportion to users' share of marginal enrollees. Appendix B.2 develops both.

5 The Regulator's Choices

We now turn the incidence results into a regulator's problem. The regulator chooses an instrument, a net price ceiling or a copay cap, and the drugs to which it applies. We consider three objectives: aggregate consumer surplus $CS_A = \delta CS_D + (1-\delta)CS_N$, net drug spending $\Pi_M + mQ$ (payments to the manufacturer), and total surplus $TS = CS_A + \Pi_I + \Pi_M$. Since the objectives differ only in how they weight the three parties, the rankings of instruments and drugs are fully determined by incidence. Proposition 7 delivers the regulator's choices.

Assumption 1 (No cash market, no valuation shift)**.** *There is no cash market ($D_0 \equiv 0$), and users' enrollment responds to the premium but not directly to the copay ($\partial s_D/\partial c = 0$).*

Let $S = \delta s_D + (1-\delta)s_N$ denote total enrollment, $S' \equiv \delta s'_D + (1-\delta)s'_N < 0$ its response to the premium, $Q = \delta s_D D(c)$ insured fills, and $X \equiv \delta s_D|D'(c)|$ the fills induced by a \$1 copay reduction. By the definition of ω_D, $\delta s'_D = \omega_D S'$ and $(1-\delta)s'_N = (1-\omega_D)S'$, so the surplus per

dollar of premium from marginal enrollment (Proposition 6) is

$$\Lambda \equiv |S'|\big[P + \omega_D(c-m)D(c)\big].$$

Proposition 7 (Incidence of marginal regulation)**.** *Under Assumption 1, to first order:*

(a) Net price ceiling. *A binding ceiling that lowers the net price by \$1 per fill lowers the premium by $\rho\,\omega_D D(c)$ and changes payoffs by*

$$\Delta CS_A = S\rho\,\omega_D D(c), \qquad \Delta\Pi_M = -Q + |S'|\,\rho\,\omega_D^2 D(c)^2(k-m),$$
$$\Delta\Pi_I = Q, \qquad \Delta TS = \Lambda\,\rho\,\omega_D D(c).$$

Users receive a share $\delta s_D/S$ of the consumer gain and non-users the remainder.

(b) Copay cap. *A \$1 copay reduction lowers the net price by $\kappa \equiv dk^*/dc$ and raises the premium by*

$$\Delta P = \rho\,\omega_D\big[(1-\kappa)D(c) + (k-c)|D'(c)|\big]. \tag{23}$$

It changes payoffs by

$$\Delta CS_A = Q - S\,\Delta P, \qquad \Delta\Pi_M = -\kappa Q + (k-m)X - |S'|\,\omega_D\,\Delta P\,(k-m)D(c),$$
$$\Delta\Pi_I = -(1-\kappa)Q - (k-c)X, \quad \Delta TS = (c-m)X - \Lambda\,\Delta P.$$

Users gain $Q - \delta s_D \Delta P$, and non-users lose $(1-\delta)s_N\Delta P$.

(c) Targeting. *A net price ceiling at a fraction θ of the list price reduces net spending by $G(1 - r - \theta)^+$, where $G = p_L Q$ is gross spending at list prices and r the rebate rate, less spending on the fills induced by the lower premium, which is of order ω_D^2. A copay cap at $\bar{c} < c$ raises consumer surplus by*

$$\int_{\bar{c}}^{c} \Big[\delta s_D(q)D(q) + S(q)\,\frac{dP^*}{dq}\Big]dq, \tag{24}$$

users' direct gain net of the premium increase paid by all enrollees ($dP^*/dq \le 0$).

If users are inframarginal, then $\omega_D = 0$ and

$$\kappa = (1-\eta)\Big[1 + \epsilon_c\,\frac{B}{c\,D(c)}\Big], \tag{25}$$

where $B \equiv (SP - \Pi_I^0 + \varphi)/(\delta s_D)$ and $\epsilon_c \equiv -cD'(c)/D(c)$ is the demand elasticity at the copay. The premium does not respond to either instrument, the ceiling transfers Q per dollar from the manufacturer to the insurer and leaves consumers and total surplus unchanged, a copay reduction

gives consumers Q and total surplus $(c-m)X$, and (24) reduces to

$$\delta s_D \int_{\bar{c}}^{c} D(q)\,dq = Q\,(c-\bar{c}) + \delta s_D \int_{\bar{c}}^{c} \big[D(q) - D(c)\big]dq.$$

For $\omega_D > 0$, κ has no closed form and is computed numerically.

Instrument choice. Consider two regulators, one that maximizes consumer welfare, and the other minimizes spending. A consumer-welfare regulator prefers the copay cap when users are inframarginal. A copay cut passes through each dollar of price reduction to consumers, minus the premium increase $S\Delta P$ it induces, while a net price ceiling passes to consumers $S\rho\,\omega_D D(c)$ per dollar of net price (Proposition 7). As users become marginal, the gap narrows: at $\omega_D = 0$, the copay passes to consumers the full dollar and the ceiling passes nothing. The net price ceiling's gain to consumers rises with premium pass-through and with users' share of marginal enrollees, while part of the copay cap's gain is clawed back through the premium at non-users' expense.

A spending-minimizing regulator can use either instrument. A ceiling lowers spending by Q per dollar, less a term of order ω_D^2 from the fills induced by the lower premium. A copay cap lowers spending by $\kappa Q - k\,\delta s_D |D'(c)|$ per dollar, which is positive unless utilization is elastic relative to the share of the copay cut passed upstream. When users are marginal, the users pushed out by the higher premium further lower spending. Market structure enters through η, which divides a copay cut between insurer and manufacturer, and through B, the insurer's gain from agreement per user. With inframarginal users, when $\epsilon_c B/(cD) > \eta/(1-\eta)$, then $\kappa > 1$ and the insurer gains from a copay cut: the manufacturer concedes more on the net price than the insurer loses in copay revenue.

Which drugs. With inframarginal users, Equation (24) states that, a consumer-welfare regulator using a copay cap should target drugs with high consumer price exposure $c - \bar{c}$ above the copay cap, many insured fills Q, and a large utilization response. The regulator should not target drugs by list price. For drugs with marginal users, part of a copay cut's gain is mitigated through a premium increase, and a net price ceiling also delivers consumer gains and may be the preferred instrument.

A spending regulator using a ceiling should target drugs with high gross spending *and* shallow rebates, since savings are $G(1-r-\theta)^+$. A drug with the highest gross spending but a rebate deeper than $1-\theta$ yields no savings at all. Medicare's selection rule ranks eligible drugs by gross spending (Cubanski & Neuman, 2026), and so omits the rebate term. When users are inframarginal, the savings from a net price ceiling are $G(1-r-\theta)^+$. With marginal users,

savings are reduced by the additional spending on the fills induced by the lower premium, a second-order effect.

Lastly, a total-surplus regulator values neither transfer. With inframarginal users, it is indifferent to net price ceilings and targets copay caps where the utilization response is large and copays are well above marginal cost, since each dollar of copay reduction raises total surplus by $(c - m)\,\delta s_D |D'(c)|$. Capping below marginal cost lowers total surplus, so the copay cap should not fall below m. With marginal users, a ceiling raises total surplus through the enrollment its premium reduction induces, while a copay cut creates a premium distortion that can make the total-surplus-maximizing copay exceed marginal cost (Proposition 6).

6 Application: Insulin in Medicare Part D

6.1 Insulin

Insulin is essential for patients with type 1 diabetes and many patients with type 2 diabetes, with no therapeutic substitute for those who require it.

Products. Insulin products differ along two dimensions. First, products differ in *delivery method*: insulin is administered either via vial-and-syringe or via prefilled pen. The two formats are imperfect substitutes, and patients often consult a provider to determine the format that best suits their lifestyle, comfort, and clinical requirements. Second, products differ in *pharmacokinetic profile*, including the onset, peak time, and duration. Long-acting (basal) insulins provide steady background glucose control and are typically taken once or twice daily, while rapid-acting (bolus) insulins manage mealtime glucose spikes. Providers prescribe the best insulin or combination of insulins for a patient based on their lifestyle, food consumption, age, and other characteristics. In the demand model below, delivery method defines the nests and length of action defines the choice set.

Market structure. The U.S. insulin market is highly concentrated (Knox, 2020). Three firms, Eli Lilly, Novo Nordisk, and Sanofi, account for the vast majority of insulin sales, including 90% of global insulin sales (Knox, 2020). Several "follow-on" biologic insulins have entered the market since 2020, but the presence of biosimilars has been limited. The prices of insulin products rose steadily through the 2010s, with the average price paid for insulin by patients with type 1 diabetes doubling from 2012 to 2016 (Knox, 2020).

Patient costs. Despite its clinical necessity, one in four diabetic patients ration their insulin, which can lead to higher hospitalization rates and direct medical costs (Aroda et al., 2024;

Knox, 2020). On average, type 1 diabetes-related medical costs total $800 per month per individual, and around 15% of insulin-dependent individuals allocate almost half of their family income to insulin prescriptions (Giannouchos et al., 2024).

6.2 Data

We use a 1% sample of 2022 Medicare beneficiaries linked to Part D prescription drug event data, the Master Beneficiary Summary File, plan characteristics, and Part D formulary files. The formulary files report each plan's tier placement and cost sharing for each drug. We focus on beneficiaries without the low-income subsidy or dual eligibility, whose cost sharing is set by the plan rather than subsidized to a few dollars.

6.3 Descriptive Evidence

6.3.1 Beneficiaries

Appendix Table A9 shows the characteristics of Medicare beneficiaries in our sample. Around 15% of individuals had any medication fill during the time period, and around 1% had an insulin prescription fill during the time period. Since medication nonadherence is an issue for diabetics, we also create a group of "likely diabetics" based on drug fill history, as summarized in Appendix Table A10.

6.3.2 Insulin Fill

Table 2 shows the characteristics of insulin fills. The market is dominated by branded products and concentrated: out of the 114 products, the top three products accounted for 38% of the market share. There is also variation in the prices of insulin products: while the average patient out-of-pocket price is $31, 10% of fills required a patient out-of-pocket payment of more than $100. Similarly, while the average list price is around $665, around 10% of fills had a list price of more than $1,500. Because the $35 copay cap exceeds the mean out-of-pocket price, it binds only for the upper tail of fills.

Appendix Table A11 shows what type of products insulin users filled in a given month, limiting to patients with an insulin fill at some point in the sample. Rapid- and long-acting insulin are most likely combined with other lengths of action (and recommended by doctors for patients without an insulin pump): 92.4% of patients with multiple length-of-action fills had a rapid insulin, and 95.6% had a long-acting insulin.

6.3.3 Insurance Coverage

We filter to stand-alone Part D plans (PDPs), so that we do not have to model insurers internalizing potential cost savings from fewer adverse events. Appendix Table A12 shows descriptive

Table 2: Insulin Fill Summary Statistics

Characteristic	Average
Share branded (vs generic)	0.95
Product market shares (top 3)	
Product A	0.19
Product B	0.11
Product C	0.08
Prices ($)	
Patient out-of-pocket	30.9
Share above $30	0.32
Share above $50	0.21
Share above $100	0.10
List price	664.5
Share above $600	0.33
Share above $1000	0.19
Share above $1500	0.10

statistics for the 776 PDPs offered across Part D regions.[3] There are around 20 PDPs available in each region, with an average premium of $48 per month. Plans either have $480 in deductible (defined standard benefit) or $0 (enhanced alternative). The average PDP enrolls around 50 individuals in our sample and takes up 1% of the market. Among likely diabetics, the average plan enrolls around 5% of patients in the region.

Appendix Figure A3 shows the share of drugs covered in some way by PDPs. On average, a plan covers slightly more than half of all insulin drugs. Appendix Figure A4 shows the distribution of plan generosity by share of a drug's list price that the enrollee is responsible for. On average, the patient is responsible for slightly more than half of the drug's list price.

6.3.4 Formulary

There are 6 different formulary tiers in the dataset. Appendix Figure A5 shows that there is substantial variation in which tier drugs are placed in, even within type and length-of-action groups. Most of the drugs are placed most frequently in tier 3, while some drugs are often placed in tier 1.

Appendix Figures A6 and A7 show the distribution of list price and patient out-of-pocket price by a drug's most often placed tier. Drugs most often placed in higher tiers typically have higher list prices and higher out-of-pocket costs. This formulary-driven variation in out-of-pocket prices for the same product is what identifies demand.

[3] Among non-subsidized, non-dual enrollment enrollees only.

6.4 Insulin Demand

We define a market by the combination of a time period t (quarter) and a demographic group g (defined as age bins: below 70, between 70-80, and above 80). Within each market, patients choose to fill a prescription for product j from the set of available insulin products J_t (including basal, bolus, and premix options) or choose the outside option of no fill.

Sample. We focus on non-subsidized individuals in stand-alone Part D plans (including employer-sponsored plans) who are likely to be diabetic based on their prescription history.[4] For drug products, we include products that have at least 1% market share in a length-of-action category. For each drug product, we construct an out-of-pocket cost using plans' pre-initial coverage limit standard in-network copay or coinsurance associated with that tier. If the cost sharing for in-network pharmacies is not reported, we fill in using the cost sharing for in-network preferred pharmacies, followed by in-network non-preferred pharmacies. If a drug is covered by the Senior Savings model, its out-of-pocket cost is set as $35 per month. Beneficiaries who filled two products in the same length-of-action category in a quarter (about 10% of insulin users) are excluded.

Logit demand. Conditional on health status, the indirect utility that a patient in group g receives from choosing product j in length of action l in quarter t is given by:

$$u_{jgt} = \alpha_g \text{OOP}_{jgt} + \xi_{jt} + \varepsilon_{jgt}$$

where OOP_{jgt} is the out-of-pocket price faced by the patient; α_g is the price sensitivity parameter, which we allow to vary by demographic group (e.g., $\alpha_g = \alpha_0 + \alpha_1 \mathbb{I}(\text{Age} > 70) + \alpha_2 \mathbb{I}(\text{Age} > 80)$); ξ_{jt} are product-time fixed effects capturing product-specific quality (brand reputation) and time-varying trends in insulin usage; and ε_{jgt} is an idiosyncratic error term assumed to be i.i.d. Type I Extreme Value.

We also include a nested logit specification to account for the high degree of substitutability among products with the same delivery mechanism (e.g., patients are more likely to switch between two different insulin pens than from a pen to a vial): we organize the set of products J_t into nests based on **form** (Pen vs. Vial). We assume the error term ε_{jgt} follows a generalized extreme value (GEV) distribution, allowing for correlation in unobserved utility among products within the same nest. This structure yields the following linear estimating equation for

[4] We limit the sample to non-subsidized enrollees (no dual enrollment, no low income subsidies), since the cost-sharing level among subsidized enrollees is extremely low (quarterly average of $4 for insulin products, compared to $96 for non-subsidized enrollees).

the demand of demographic group g (Berry, 1994):

$$\ln(s_{jgt}) - \ln(s_{0gt}) = \alpha_g \text{OOP}_{jgt} + \sigma \ln(s_{j|nest,g,t}) + \xi_{jt}$$

where s_{jgt}/s_{0gt} is the ratio of the market share of product j to the share of the outside option (no fill) for demographic group g; $s_{j|nest,g,t}$ is the *conditional market share* of product j within its nest (e.g., the share of Lantus Solostar among all Pen fills in that market); and σ is the nesting parameter, where $0 \leq \sigma < 1$ measures the correlation of utility within the nest. As $\sigma \to 1$, products within the same nest become close substitutes; if $\sigma = 0$, the model collapses to the standard logit. This specification allows us to recover the own-price elasticity for product j as

$$\eta_{jgt} = \frac{\alpha_g \text{OOP}_{jgt}}{1-\sigma} \left[1 - \sigma s_{j|nest,g,t} - (1-\sigma) s_{jgt}\right], \tag{26}$$

reflecting that consumers are more price-sensitive when switching between products within the same nest.

Identification. The variation in OOP_{jgt} is primarily driven by the Pharmacy Benefit Manager's (PBM) formulary design: drugs are assigned to distinct tiers, which determine the copayment or coinsurance rate, and because formulary placement varies across plans and years due to rebate negotiations, we observe substantial price variation for the same product across different markets, identifying α_g.

Results. Table 3 shows the results. Price coefficients are negative and precisely estimated. The nesting parameter is large (0.82), indicating strong substitution within delivery forms. The median product-level own-price elasticity is -0.16 to -0.20 in the plain logit and -0.71 in the nested logit.

A product's own-price elasticity combines switching among insulin products with changes in whether a patient fills at all, and only the latter is what a class-wide copay cap affects; with $\sigma \approx 0.8$, most of the product-level response is within-nest switching. We therefore do not calibrate aggregate utilization to -0.71 but use the quasi-experimental aggregate response in Fact 2, an arc elasticity of about 0.16 among highly exposed patients and roughly zero on average. Our baseline aggregate elasticity is 0.10, and we report 0.05 and 0.16 as alternatives.

6.5 Calibration

We calibrate the full model at the monthly frequency with one composite insulin product. Table 4 lists parameters and targets. The user share is $\delta = 0.05$, the share of Part D enrollees with a potential insulin fill. Drug demand is logit, $D(q) = \Lambda(a(v-q))$, with a and v chosen

Table 3: Drug Demand Estimates

	(1)	(2)	(3)
Price Sensitivity (α, 00s)	-0.165^{***}	-0.154^{***}	-0.095^{***}
	(0.006)	(0.009)	(0.006)
Age>70		-0.028^{**}	
		(0.010)	
Age>80		0.001	
		(0.012)	
Demographics			
Age>70		-0.358^{***}	
		(0.045)	
Age>80		-0.069	
		(0.058)	
Dissimilarity Parameter			0.823^{***}
Median own price elasticity	-0.194	-0.162	-0.705
Observations	3,788	4,137	3,788
R^2	0.29	0.34	0.44

to match the observed fill rate $D(30.90) = 0.56$ and an aggregate out-of-pocket elasticity of 0.10. The implied value of insured access, the logit inclusive value, is $W(30.90) = \$112$ per month. In the baseline there is no cash market: our claims data do not identify purchases outside Part D, so we assume users without coverage do not buy at the list price and relax this assumption below. The two insurance parameters (λ, g) are chosen so that the equilibrium premium equals the mean PDP premium of \$48 and a uniform per-enrollee cost is passed through to the premium at rate 0.45, the average of Cabral et al. (2018). The bargaining weight is $\eta = 0.5$, and marginal cost is zero. Rebates are not observed in our data, so the net price is an output of the calibration rather than a target.

Benchmarking the net price. The calibrated rebate of 94.9% is deeper than external estimates: commercial discounts on the four insulins most used in Part D reached 60.5% of gross sales in 2019 (Dickson, Gabriel, Gellad, & Hernandez, 2023), Senate Finance Committee documents show Part D rebates of 68% to 80% of list in 2018–19 (U.S. Senate Committee on Finance, 2021), and because a maximum fair price cannot exceed the prior net price, NovoLog's 2026 price bounds its prior rebate at 76%. The baseline rebate is too deep because the one-drug model gives the insurer only the premium surplus from covering insulin; φ captures what it omits. Calibrating φ so that the rebate is 70% (a net price of \$199.35) requires $\varphi = \$9.24$ per beneficiary-month, about 19% of the premium. Table 7 and Appendix Table A5 carry this specification through every exercise: the incidence shares do not change, only the dollar size

Table 4: Calibration

Parameter	Value	Source or target
User share δ	0.05	Share of Part D enrollees with a potential insulin fill
Copay c_0	\$30.90	Mean OOP per insulin fill
List price p_L	\$664.50	Mean list price per insulin fill
Fill probability $D(c_0)$	0.56	Share of user-months with an insulin fill
Aggregate OOP elasticity	0.10	Myerson et al. (2026): 0 on average, 0.16 in top OOP decile
Drug demand (a, v)	(0.0074, 63.7)	Implied by the two targets above
Premium sensitivity λ	0.0502	Premium \$48 (mean PDP premium) and pass-through 0.45 (Cabral et al., 2018)
Other coverage value g	\$51.7	
Insurer bargaining weight η	0.5	Symmetric Nash; varied 0.2–0.8
Marginal cost m	0	Varied to \$5 per fill
Insurer's disagreement loss φ	0	Varied: calibrated to a 70% rebate
Cash market	None	Varied: 2% cash purchase at list, elasticity 0.5 or 1.5
Implied baseline equilibrium		
Net price k^* (rebate rate)	\$33.98 (94.9%)	
Enrollment: users, non-users	99.7%, 54.6%	
Users' share of marginal enrollees ω_D	0.06%	Proposition 2

of what a price ceiling can transfer (\$16.70 rather than \$2.85 per beneficiary-year at 75% of the net price) and the slope κ, which rises to 1.11. We keep $\varphi = 0$ as the baseline so that the net price is a model output rather than a target.

6.6 The \$35 Copay Cap

In this section, we first simulate the potential effects of a \$35 copay cap that applies to the single plan in the market, with a single copay and variation in copays.

Single composite copay. We first trace the effects from changing a single composite copay from \$30.90 to zero, holding the list price fixed, to measure the responses the theory characterizes. We then simulate groups of patients with different copays, for some of whom the copay cap binds, to consider heterogeneous effects.

The results from changing the composite copay from \$30.90 to zero are shown in Appendix Table A6 and match the theoretical predictions from the model. Each dollar of copay removed lowers the bargained net price by \$0.558, against 0.560 predicted by equation (25). Since users are inframarginal ($\omega_D = 0.06\%$), the premium moves by less than a cent, and the insurer and the manufacturer each bears half of the cost of the copay decrease, \$5.15 per beneficiary-year

at full elimination, the $1-\eta$ split of Lemma 1. Users' surplus rises by $217 per user-year and total surplus by $0.49 per beneficiary-year, one-twentieth of the transfer. The model thus predicts that a copay cap raises rebates, as the Senior Savings evaluation found.

That exercise isolates the mechanisms but is not representative of what happened in reality. Medicare's copay cap is $35, above the mean out-of-pocket price of $30.90, so it binds only in the upper tail of cost sharing, where Fact 2 locates its effects.

Variation in copays. To simulate the real-world change, we consider the effects of different copay caps when users face different existing copays. We create four bins of out-of-pocket price per fill : 65% of users pay $5.59 per fill, and the rest pay $40, $70 and $150 (roughly a quarter of the mean list price), with user weights chosen to match the shares of fills above $30, $50 and $100 and the fill-weighted mean copay observed in the data.

In this exercise, all users have the same drug demand function, whose two parameters are set so that the mean fill rate is 0.56 and the response of the most exposed type matches the top out-of-pocket decile in Myerson et al. (2026), where the copay cap lowered out-of-pocket cost from about $91 to $45 per supply and raised fills by 8%. The implied aggregate elasticity at the mean copay is 0.05. One net price is bargained for all types, and (λ, g) are recalibrated to the premium and pass-through targets. The baseline reproduces the single-type equilibrium closely (net price $34.02, rebate 94.9%).

Table 5: Cost-Sharing Caps with Heterogeneous Copays

	Users	Mean	Δ Fills (%)		Net	User gain ($/yr)		Non-user	Δ per beneficiary-year ($)		
Policy	affected (%)	OOP ($)	All	Affected	price ($)	All	Affected	($/yr)	Insurer	Mfr.	Total
Baseline	0	30.90	0.0	0.0	34.02	0	0	0.00	0.00	0.00	0.00
Copay cap at $100	12	26.61	1.0	9.7	31.70	35	288	0.00	-0.68	-0.68	0.41
Copay cap at $50	24	19.40	2.3	11.0	27.86	89	372	0.00	-1.86	-1.86	0.72
Copay cap at $35 (2023)	35	15.57	3.0	9.2	25.83	116	332	0.00	-2.50	-2.50	0.81
Copay cap at $25	35	12.25	3.5	10.9	24.07	140	400	0.00	-3.06	-3.06	0.87
Copay eliminated	100	0.00	5.4	5.4	17.62	226	226	0.00	-5.19	-5.19	0.93
Ceiling alone, 22% below net	0	30.90	0.0	0.0	26.54	0	0	0.00	2.51	-2.51	0.00
Ceiling with linked copay cap (2026)	35	5.95	4.5	14.0	20.75	184	527	0.00	-4.15	-4.15	0.92

Notes: Users have copays of $5.59, $40, $70 and $150 per fill in proportions matching Table 2; one net price is bargained for all. "Affected" users are those whose copay exceeds the copay cap. Mean out-of-pocket price is per insured fill. User gains are per user-year, averaged over all users and over affected users. The ceiling in the last two rows is 22% below the bargained net price ($26.54); the linked copay cap is the lesser of $35 and 25% of the ceiling, $6.63. Changes are relative to the baseline.

Table 5 reports the results for copay caps at $100, $50, $35, $25 and zero. The $35 copay cap binds for 35% of users, who account for 32% of fills. It lowers the mean out-of-pocket price per fill by half, raises fills by 3.0% overall and by 9.2% among affected users. This follows the pattern of Fact 2: a small aggregate utilization response that is concentrated among the

exposed. The tail response is a calibration target; the aggregate response is a prediction, consistent with the absence of an average effect in Myerson et al. (2026). Our mean out-of-pocket reduction exceeds the 21% they estimate because the non-subsidized PDP fills in our sample have a heavier upper tail than their full sample. Affected users gain $332 per year, $116 averaged over all users. The cost falls on the insurer and the manufacturer in equal shares, $2.50 each per beneficiary-year, with the net price falling by $8.19 per fill. Since users are inframarginal, non-users do not bear the cost of the copay caps. Relative to eliminating the copay ($226 per user-year at a cost of $5.19 to each firm), the copay cap delivers about half the transfer at half the cost and directs it to the users with the highest exposure; a $100 copay cap reaches 12% of users but gives each of them $288 per year, the targeting rule of Proposition 7(c).

Alternative copay policies. The last two rows in Table 5 combine the two instruments as the 2026 rule does (CMS, 2025): a net price ceiling 22% below the bargained net price, the average net reduction in the first negotiation cycle, with cost sharing capped at the lesser of $35 and 25% of the ceiling. The ceiling alone transfers $2.51 per beneficiary-year from the manufacturer to the insurer and nothing to users. Linking cost sharing to it lowers the copay cap to $6.63, which binds for the same 35% of users; they gain $527 per year, and the net price falls to $20.75, below the ceiling, so the ceiling itself does not bind. Linking the copay to the regulated price is what moves the manufacturer's concession from the insurer to patients.

6.7 Net Price Ceilings

In this section, we simulate the effects of regulating the net price ceiling. We also consider the effects of targeting the list price, with and without a cash market. Table 6 reports the results.

Net price ceilings. Panel A sets the ceiling at 90%, 75% and 50% of the bargained net price of $33.98. As Proposition 3 predicts, each ceiling is a transfer from the manufacturer to the insurer. When the ceiling is set to 75% of the bargained net price, the manufacturer loses $2.85 per beneficiary-year and the insurer gains $2.85, while consumers gain $0.01 and total surplus changes by $0.01. The insurer keeps the saving for the same reason it absorbs a copay cut: under community rating, a cost change concentrated on inframarginal users barely moves the premium. The transfer scales one for one with the depth of the ceiling.

List price ceilings. While we focus on net price regulations, we consider the impact of regulating list prices. Panels B and C consider list price regulations, with and without a cash

Table 6: Counterfactual Net Price Ceilings and List-Price Cuts

	Net			Fills	Δ Consumer surplus			Δ per beneficiary-year		
Policy price	price ($)	Rebate (%)	Premium ($/mo)	per user (/mo)	User ($/yr)	Non-user ($/yr)	All ($/yr)	Insurer ($)	Mfr. ($)	Total ($)
A. Net price ceiling (90%, 75%, 50% of the bargained net price)										
$30.58	30.58	95.4	48.00	0.558	0.0	0.00	0.00	1.14	-1.14	0.00
$25.48	25.48	96.2	48.00	0.558	0.0	0.01	0.01	2.85	-2.85	0.01
$16.99	16.99	97.4	48.00	0.558	0.0	0.02	0.02	5.69	-5.69	0.02
B. List-price cut, no cash market										
$500.00	33.98	93.2	48.00	0.558	0.0	0.00	0.00	0.00	0.00	0.00
$200.00	33.98	83.0	48.00	0.558	0.0	0.00	0.00	0.00	0.00	0.00
$100.00	33.98	66.0	48.00	0.558	0.0	0.00	0.00	0.00	0.00	0.00
$25.48	25.48	0.0	48.00	0.558	0.0	0.01	0.01	2.85	-2.85	0.01
C. List-price cut, cash market (2% purchase at list, elasticity 0.5)										
$500.00	43.88	91.2	47.99	0.553	0.7	0.09	0.12	0.55	-0.63	0.03
$200.00	38.67	80.7	47.95	0.549	2.7	0.35	0.47	2.11	-2.46	0.12
$100.00	36.26	63.7	47.93	0.548	3.7	0.49	0.65	2.82	-3.32	0.15

Notes: Copay fixed at $30.90; baseline list price $664.50 and bargained net price $33.98. Panel A caps the price the insurer pays; the first column is the ceiling in panel A and the new list price in panels B and C. Panels B and C cut the list price, as insulin manufacturers did in 2023–24; the last row of panel B cuts it to 75% of the bargained net price, where it coincides with the second row of panel A. Panel C adds a cash market in which 2% of users buy at the baseline list price with elasticity 0.5, recalibrated to the same premium and pass-through targets; changes are measured against each panel's own baseline.

market.[5] In panel B, list-price cuts to $500, $200 and $100 change nothing when there is no cash market: the rebate rate falls from 94.9% to accommodate the lower list price, and the net price, premium, enrollment and all surpluses are unchanged. This result follow Proposition 4. A cut to 75% of the bargained net price reproduces the second row of panel A exactly, which is part (iii) of that proposition. In panel C, 2% of users would buy at the list price if the drug were not covered. The list price now moves the manufacturer's outside option, and a cut to $200 lowers the bargained net price from $45.69 to $38.67, costs the manufacturer $2.46 per beneficiary-year, and raises users' surplus by $2.67 per user-year, mostly by lowering the price paid by cash buyers. The results illustrate that list price regulations are only effective for settings with a cash market, since it changes the prices faced by cash buyers.

6.8 Incidence and Market Structure

Appendix B.3 decomposes each policy into the direct, premium, and bargaining channels of equation (8). Table 7 shows how the incidence of the two instruments varies with market structure, re-calibrating the insurance parameters in each row to the same premium and the

[5] Appendix Figure A2 traces the bargained net price against the list price and against a net price ceiling, with and without a cash market

stated pass-through. For the copay cut it reports the shares of the cost borne by the insurer, the manufacturer, and non-users; for a ceiling at 75% of the bargained net price, the gains of the insurer and of consumers per dollar of manufacturer loss.

Table 7: Incidence by Market Structure

		Copay from \$30.90 to \$0				Net price ceiling	
	ω_D	User gain	Share of cost borne by (%)			Gain per \$1 of mfr. loss (%)	
Market structure	(%)	(\$/yr)	Insurer	Mfr.	Non-users	Insurer	Consumers
Baseline	0.06	217	50	50	1	100	0
Weak insurer bargaining, $\eta = 0.2$	0.06	217	20	80	1	100	0
Strong insurer bargaining, $\eta = 0.8$	0.06	217	79	20	1	100	0
Low pass-through, 0.13	0.00	218	50	50	0	100	0
High pass-through, 0.74	1.18	206	45	45	11	100	6
Users more marginal	3.37	179	28	28	44	100	21
Larger gains from agreement	0.06	218	50	50	0	100	0

Notes: Each row re-calibrates (λ, g) to a \$48 premium and the stated pass-through (0.45 unless noted). Cost of the copay cut is the sum of insurer and manufacturer losses and non-users' premium losses, per beneficiary-year. The ceiling is set at 75% of each specification's bargained net price. "Users more marginal" sets the value of insulin coverage to \$25 per month, so that more users are near the enrollment margin. "Larger gains from agreement" sets the insurer's disagreement loss φ so that the baseline rebate is 70% (net price \$199.35). Appendix Table A5 reports further specifications.

Three patterns stand out. First, bargaining power divides the cost of a copay cut between the firms. The manufacturer bears 20% when the insurer is strong ($\eta = 0.8$) and 80% when it is weak ($\eta = 0.2$), as equation (25) predicts.

Second, non-users pay only when users are marginal enrollees. The share of cost they bear is about 1% at the baseline, 11% with high premium pass-through, and 44% when users value coverage less and so are more often marginal. Third, a net price ceiling is a transfer to the insurer, and the consumer receives very little of the savings. Consumers receive 0.3 cents per dollar of manufacturer loss at the baseline, 6.0 cents with high pass-through, and 21.1 cents when users are more marginal. The last row raises the insurer's gains from agreement until the rebate matches external evidence; the shares in every column are unchanged, and only the dollar amounts behind them scale with the net price.

Appendix Table A5 repeats the exercise for alternative elasticities, user shares, marginal cost, and cash markets; total surplus falls only where users are most marginal or pass-through is highest, the premium-distortion case of Proposition 6.

6.9 Competing Plans: A Plan-Level versus a Market-Wide Copay Cap

The single-insurer model in Section 6.6 predicts that non-users bear almost none of the costs from a copay cut, because insulin users are inframarginal to coverage, yet Fact 3 shows that

non-users in Senior Savings PDPs paid $30–$65 more per year through premiums. The difference is that the Senior Savings copay cap was adopted plan by plan.

To illustrate this argument, we extend the application to n symmetric plans that compete in premiums. Users and non-users choose among the n plans, with no coverage as the outside option, with logit preferences. All plans are represented by one pharmacy benefit manager, who negotiates a single net price, where no plan covers the drug in disagreement. Each plan sets its premium as a best response to the others' premiums. We hold the drug demand at its baseline and re-calibrate (λ, g) to the $48 premium and the baseline non-user enrollment rate of 54.6%; a cost common to all plans then passes through at a rate close to one (1.00 at $n = 20$), a property of logit competition among small plans rather than a calibration choice.

Competition among plans changes which enrollees are marginal: with one insurer the only margin is coverage itself, to which users are inframarginal ($\omega_D = 0.06\%$). With $n = 20$ plans, each plan's marginal enrollee is a potential enrollee in another plan, and users' share of a plan's marginal enrollees is 8.1% even though 93% of them hold coverage. Table 8 compares a market-wide mandate that sets the copay to zero in every plan, as the Inflation Reduction Act did with the $35 copay cap, with a plan-level copay cap adopted by one plan while the others keep the $30.90 copay, as in the Senior Savings Model, for $n = 5$ and $n = 20$ and next to the single-insurer results.

Three results follow. First, under the market-wide mandate, the net price falls by $17.93 per fill, as with one insurer, but the premium rises by $0.79 per month rather than $0.01, because each plan prices against its plan-level margin, at which users are marginal. Non-users lose $5.2 per year, and total surplus falls by $1.20 per beneficiary-year where it rose by $0.49 with one insurer: with near-complete pass-through, the premium distortion in Proposition 6 outweighs the gain from induced fills.

Second, a plan-level copay cap is paid by the capping plan's own pool. The plan's user enrollment rises by 38% and its non-user enrollment falls by 5.6%, its premium rises by $2.65 per month, and non-users who remain lose $32 per year against a gain of $186 for users who remain, a ratio of 0.17 (0.13 with five plans). In CMS's evaluation, non-users in participating PDPs lost $30–$65 per year, 0.08–0.20 of users' gain, and insulin-user enrollment rose by 75%–92%. The model was not fit to these moments, and the correspondence is an order-of-magnitude check rather than a test, but it reproduces a pattern that lies outside the single-insurer model.

Third, the bargaining channel is a feature of market-wide policy: under the plan-level copay cap the net price falls by only $1.34 per fill, because the capping plan is a small share of insured fills and the net price is common to all plans. The two policies regulate the same price for the same drug; what differs is the margin against which premiums are set. The extension

Table 8: Eliminating the Copay with Competing Plans: Market-Wide Mandate versus Plan-Level Copay Cap

	Single	5 plans		20 plans	
Change relative to baseline	insurer	Market-wide	One plan	Market-wide	One plan
A. Prices					
Premium of capped plan(s) ($/month)	0.01	0.71	2.13	0.79	2.65
Premium of other plans ($/month)	–	–	-0.30	–	-0.08
Net price ($/fill)	-17.07	-17.89	-5.23	-17.93	-1.34
B. Selection into the capping plan					
Users' share of marginal enrollees, before (%)	0.06	7.66	7.66	8.10	8.10
Users' share of marginal enrollees, after (%)	0.03	7.81	9.67	8.31	11.19
User enrollment (%)	0.2	1.9	34.2	2.2	37.7
Non-user enrollment (%)	0.0	-0.8	-4.8	-0.8	-5.6
C. Incidence					
Users' gain ($/user-year)	217	200	192	197	186
Non-users' loss ($/non-user-year)	0.1	4.6	25.6	5.2	31.7
Ratio of non-user loss to user gain	0.00	0.02	0.13	0.03	0.17
Insurers ($/beneficiary-year)	-5.15	-2.38	-0.61	-1.88	-0.13
Manufacturer ($/beneficiary-year)	-5.15	-4.45	-1.21	-4.23	-0.29
Total surplus ($/beneficiary-year)	0.49	-1.23	-0.63	-1.20	-0.18

Notes: Copay reduced from $30.90 to $0 in every plan ("market-wide") or in one plan only ("one plan"). The single-insurer column repeats Appendix Table A6. Plans are symmetric; (λ, g) are re-calibrated for each n to a $48 premium and 54.6% non-user enrollment; one net price is negotiated for all plans. Panel B reports users' share of the capping plan's marginal enrollees before and after the policy and the change in the plan's enrollment by group. In the one-plan columns, users' gain and non-users' loss are for enrollees who remain in the capping plan, and the insurer row sums over all plans. In the market-wide and single-insurer columns they are the average over all users and non-users, including those who change coverage.

abstracts from plan asymmetries, risk adjustment, reinsurance and premium subsidies, and PBMs serving subsets of plans.

6.10 Which Drugs Should a Regulator Target?

To understand which characteristics make a drug a good candidate for targeting, we conduct a simulation where we hold fixed the insurance market (premium sensitivity, other coverage value, bargaining weight), and vary one characteristic of the drug at a time. The value of coverage to users is held at $112 per month throughout, so that the archetypes differ only in the stated respect. Specifically, we vary the drug's existing copay, demand elasticity, list price, and user base.

We apply two policies to each drug: a $10 copay cap and a net price ceiling at 40% of the list price, roughly the average negotiated discount in Table 1. Table 9 reports the results.

Table 9: Regulating Archetypal Drugs in a Common Insurance Market

	Baseline			$10 copay cap		Ceiling at 40% of list		
Drug	Gross spending	Net spending	Rebate (%)	Δ Cons. surplus	Δ Net spending	Δ Cons. surplus	Δ Net spending	Δ TS (copay cap)
A. Baseline insulin	222.6	11.4	95	7.18	-3.37	0.00	0.00	0.44
B. High exposure ($60 copay)	222.6	16.3	93	17.35	-8.20	0.00	0.00	0.95
C. Low exposure ($12 copay)	222.6	8.2	96	0.67	-0.30	0.00	0.00	0.06
D. Elastic use (0.30)	222.6	11.4	95	7.63	-3.15	0.00	0.00	1.33
E. Shallow rebate ($70 list)	23.4	11.4	51	7.18	-3.37	0.01	-2.00	0.44
F. Many users (10%)	445.0	23.4	95	14.31	-6.72	0.00	0.00	0.87
G. Many users, shallow rebate	46.9	23.4	50	14.31	-6.72	0.02	-4.63	0.87

Notes: Dollars per beneficiary-year. Gross spending values insured fills at the list price; net spending is payments to the manufacturer. Each drug differs from the baseline insulin (copay $30.90, aggregate elasticity 0.10, list price $664.50, user share 0.05) only as stated. Premiums and rebates are re-solved in every case.

The consumer welfare ranking follows equation (24): the copay cap raises consumer surplus most for the high-exposure drug ($17.35 per beneficiary-year) and the drug with many users ($14.31), and hardly at all for the low-exposure drug ($0.67). In contrast to the copay cap, the price ceiling barely changes consumer welfare.

While the Medicare Drug Negotiation Program focuses on drugs with the highest gross spending, the results show that targeting these drugs may not yield the highest improvement in consumer welfare or cost-savings. For example, although a drug with a large user base, row (F), has the highest gross spending, the 40%-of-list ceiling does not change consumer welfare or net spending, since its 95% rebate already places its net price below the ceiling. Consistent with Proposition 7(b), the copay cap also lowers net spending, by $3.37 for the baseline drug and $8.20 for the high-exposure drug, less for the elastic drug ($3.15), where induced fills offset the lower net price.

6.11 Model Predictions and Existing Evidence

The model's predictions are consistent with the evidence from implemented policies. The $35 copay cap raises fills by 3.0% overall, consistent with the absence of an average effect under the national copay cap, while the 9.2% response of affected users matches the 8%–9% increases in the top out-of-pocket decile and in Senior Savings plans. Manufacturers finance part of a copay cut through larger rebates, as the Senior Savings evaluation found. The single-insurer model cannot reach that evaluation's non-user losses of 0.08–0.20 of users' gain even when only 61% of users enroll, but the competing-plans extension does (0.17), because participation was plan-level; the same extension predicts that the 2023 market-wide copay cap cost non-users $5 per year, a test that requires separating the copay cap from the concurrent Part D redesign. The first negotiation cycle realized 17% of its headline reduction, and the 70%–78% list-price cuts that insulin manufacturers made in 2023–24, after the removal of the Medicaid rebate cap made high list prices costly (ASPE, 2026; KFF, 2024), offer a further test of Proposition 4 once post-2022 data are available: little change in net prices or fixed-copay patients' prices, with gains concentrated among cash and coinsurance patients.

7 Conclusion

Who wins and who loses from drug price regulation depends on which link in the pricing chain the regulation targets and on the structure of the market. A net price ceiling transfers surplus from the manufacturer to the insurer; consumers share in it only to the extent that drug users are marginal to the plan, and in our calibration they receive less than a cent per dollar. A copay cap benefits drug users directly and concentrates the benefit on the most exposed: Medicare's $35 copay cap binds for a third of insulin users and raises their fills while leaving aggregate use nearly unchanged. Its cost is divided between insurer and manufacturer according to bargaining power, and it reaches non-users only when users are marginal enrollees, which depends on the policy's scope: a copay cap adopted by one of several competing plans is paid by that plan's non-users through selection, as in Medicare's voluntary insulin copay-cap model, while a market-wide mandate is paid by the firms. The list price enters these conclusions only through what patients would pay outside coverage, which is why list-price discounts overstate savings and why the maximum fair price, benchmarked to the prior net price, binds by construction.

These results determine the regulator's choices. A regulator that maximizes consumer welfare should cap copays and target drugs with high out-of-pocket prices, many users, and elastic utilization. A regulator that minimizes spending should target drugs with high spending *and* shallow rebates; gross spending, the basis of Medicare's selection rule, is informative only af-

ter netting out rebates, as the first negotiation cycle, in which list discounts implied nearly six times the net savings CMS projects, illustrates.

Several limitations affect the magnitudes. The model has one drug and one manufacturer; competition among therapeutic substitutes shapes the disagreement point (Ho & Lee, 2026) and is where rebates could survive below a binding ceiling. The competing-plans extension has symmetric plans and a single bargaining unit. We abstract from federal reinsurance and subsidies, the phases of the Part D benefit, health effects of adherence, and dynamic effects on innovation (Agha, Kim, & Li, 2022; Garber, Jones, & Romer, 2006). Premium sensitivity is disciplined by external pass-through estimates rather than estimated plan choice, rebates are unobserved so the net price is a model output, and cash purchases outside Part D are unobserved and, by Proposition 5, govern how far neutrality extends. The mechanisms, and the targeting rules they imply, are supported by the administrative evidence from Medicare's insulin copay cap and its first negotiated prices.

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A Proofs

Proof of Proposition 1. We derive the entries of Proposition 1 channel by channel. In this derivation we write $s(\cdot)$ for the demand function $D(\cdot)$ and $r = p-k$ for the rebate, and we label each term by the channel it operates through: the payment from the insurer to the manufacturer, the premium, the rebate, and consumers' out-of-pocket payments. Case 1 is the protocol in which the manufacturer makes the offer and Case 2 the protocol in which the insurer makes the offer; in Case 1 the rebate satisfies $S(k) = 0$, and in Case 2 it holds the manufacturer to $pD(p)$.

Social welfare is $W(c) + cs(c)$. In other words, it is total consumer value from the drug net of out-of-pocket payments. However, we are interested in not just social welfare, but also how it is distributed among the manufacturer, the insurer, and consumers.

There are externalities not captured in this stylized model: manufacturer welfare may have positive externalities due to innovation incentives, insurer welfare may have positive externalities because it can be passed on to enrolled consumers in other drug markets, and consumer welfare may have positive externalities because it is tied to public health.

List price. Suppose a policy regulates the list price p.

In Case 1 (where the manufacturer has all bargaining power),

$$\frac{\partial}{\partial p}\text{Manufacturer Welfare} = \underbrace{s(c)}_{\text{Payment}} + \underbrace{[-W'(p) - s(c)]}_{\text{Rebate}} = -W'(p) \tag{27}$$

$$\frac{\partial}{\partial p}\text{Insurer Welfare} = \underbrace{(-W'(p))}_{\text{Premium}} + \underbrace{(-s(c))}_{\text{Payment}} + \underbrace{[W'(p) + s(c)]}_{\text{Rebate}} = 0 \tag{28}$$

$$\frac{\partial}{\partial p}\text{Consumer Welfare} = \underbrace{W'(p)}_{\text{Premium}} \tag{29}$$

If the list price increases, the direct effect is that the insurer pays more per unit to the manufacturer. Some of that cost is passed on to consumers through higher premiums. The rebate adjusts so the insurer is as well off as before. The manufacturer is better off and consumers are worse off, but social welfare is unchanged.

In Case 2 (where the insurer has all bargaining power),

$$\frac{\partial}{\partial p}\text{Manufacturer Welfare} = \underbrace{s(c)}_{\text{Payment}} + \underbrace{[-s(c) + s(p) + ps'(p)]}_{\text{Rebate}} = s(p) + ps'(p) \tag{30}$$

$$\frac{\partial}{\partial p}\text{Insurer Welfare} = \underbrace{[-\alpha W'(p) + (1-\alpha)(s(p) + ps'(p))]}_{\text{Premium}} + \tag{31}$$

$$\underbrace{(-s(c))}_{\text{Payment}} + \underbrace{[s(c) - s(p) - ps'(p)]}_{\text{Rebate}} \tag{32}$$

$$= -\alpha W'(p) - \alpha[s(p) + ps'(p)] \tag{33}$$

$$\frac{\partial}{\partial p}\text{Consumer Welfare} = \underbrace{\alpha W'(p) - (1-\alpha)[s(p) + ps'(p)]}_{\text{Premium}} \tag{34}$$

If the list price increases, the direct effect is that the insurer pays more per unit to the manufacturer. The premium and rebate adjust. Welfare is redistributed among the manufacturer, the insurer, and consumers, but social welfare is unchanged.

Net price ceiling. Suppose a policy imposes a ceiling $\bar{k}$ on the net price.

In Case 1, the manufacturer would set $k = k^{\max}$, so a ceiling $\bar{k} < k^{\max}$ binds and $k = \bar{k}$. The insurer's welfare is $\alpha S(\bar{k})$ and consumers' is $W(p) + (1-\alpha)S(\bar{k})$, where $\partial S/\partial \bar{k} = -D(c)$. Hence

$$\frac{\partial}{\partial \bar{k}}\text{Manufacturer Welfare} = \underbrace{D(c)}_{\text{Payment}}, \tag{35}$$

$$\frac{\partial}{\partial \bar{k}}\text{Insurer Welfare} = \underbrace{-D(c)}_{\text{Payment}} + \underbrace{(1-\alpha)D(c)}_{\text{Premium}} = -\alpha D(c), \tag{36}$$

$$\frac{\partial}{\partial \bar{k}}\text{Consumer Welfare} = \underbrace{-(1-\alpha)D(c)}_{\text{Premium}}. \tag{37}$$

If the ceiling falls, the insurer pays the manufacturer less per unit and passes a share $1-\alpha$ of the saving to consumers through the premium. Total surplus is unchanged.

In Case 2, the insurer already holds the manufacturer to its outside option, $k = k^{\min}$. A ceiling $\bar{k} \geq k^{\min}$ therefore does not bind, and no payoff changes. A ceiling $\bar{k} < k^{\min}$ violates the manufacturer's participation constraint, so the drug is not covered and total surplus falls by G.

Copay. Suppose a policy regulates the copay c.

In Case 1,

$$\frac{\partial}{\partial c}\text{Manufacturer Welfare} = \underbrace{ps'(c)}_{\text{Payment}} + \underbrace{[W'(c) - ps'(c) + s(c) + cs'(c)]}_{\text{Rebate}} \tag{38}$$

$$= W'(c) + s(c) + cs'(c) \tag{39}$$

$$\frac{\partial}{\partial c}\text{Insurer Welfare} = \underbrace{W'(c)}_{\text{Premium}} + \underbrace{(-ps'(c))}_{\text{Payment}} + \underbrace{[s(c) + cs'(c)]}_{\text{OOP}} + \tag{40}$$

$$\underbrace{[-W'(c) + ps'(c) - s(c) - cs'(c)]}_{\text{Rebate}} \tag{41}$$

$$= 0 \tag{42}$$

$$\frac{\partial}{\partial c}\text{Consumer Welfare} = \underbrace{W'(c)}_{\text{Value}} + \underbrace{(-W'(c))}_{\text{Premium}} = 0 \tag{43}$$

If the copay increases, the direct effects are that consumers lose value from the drug, out-of-pocket payments from consumers to the insurer are shifted, and the insurer buys fewer units from the manufacturer. The premium is reduced so consumers are as well off as before. The rebate adjusts so the insurer is as well off as before. All of the social welfare change is borne by the manufacturer.

In Case 2,

$$\frac{\partial}{\partial c}\text{Manufacturer Welfare} = \underbrace{ps'(c)}_{\text{Payment}} + \underbrace{(-ps'(c))}_{\text{Rebate}} = 0 \tag{44}$$

$$\frac{\partial}{\partial c}\text{Insurer Welfare} = \underbrace{[\alpha W'(c) - (1-\alpha)(s(c) + cs'(c))]}_{\text{Premium}} + \tag{45}$$

$$\underbrace{(-ps'(c))}_{\text{Payment}} + \underbrace{[s(c) + cs'(c)]}_{\text{OOP}} + \underbrace{ps'(c)}_{\text{Rebate}} \tag{46}$$

$$= \alpha[W'(c) + s(c) + cs'(c)] \tag{47}$$

$$\frac{\partial}{\partial c}\text{Consumer Welfare} = \underbrace{W'(c)}_{\text{Value}} + \underbrace{[-\alpha W'(c) + (1-\alpha)(s(c) + cs'(c))]}_{\text{Premium}} \tag{48}$$

$$= (1-\alpha)[W'(c) + s(c) + cs'(c)] \tag{49}$$

If the copay increases, the direct effects are that consumers lose value from the drug, out-of-pocket payments from consumers to the insurer are shifted, and the insurer buys fewer units from the manufacturer. The premium adjusts to redistribute welfare between the insurer and consumers. The rebate increases so the manufacturer is as well off as before. All of the social welfare change is distributed between the insurer and consumers.

Proof of Lemma 1. Write $G_M = \Pi_M - \Pi_M^0 = \delta s_D[(k-m)D(c) - (p_L - m)D_0]$ and $G_I = \Pi_I - \Pi_I^0 + \varphi$. By the envelope theorem applied to the insurer's premium choice, $dG_I/dk = -\delta s_D D(c) = -Q$. With $\partial P^*/\partial k = 0$, s_D does not vary with k, so $dG_M/dk = Q$. The first-order condition $(1-\eta)Q/G_M = \eta Q/G_I$ gives $G_I = \frac{\eta}{1-\eta} G_M$, hence $G_M = (1-\eta)J$ where $J = G_M + G_I$. Substituting for G_M and solving for k gives (12). Single-peakedness implies that if $k^u > \bar{k}$ the constrained optimum is the boundary $k = \bar{k}$. If $\partial P^*/\partial k \neq 0$, then $dG_M/dk = Q(1+\varphi)$ with $\varphi = \delta s_D' (\partial P^*/\partial k)[(k-m)D(c) - (p_L - m)D_0]/Q$, and the same steps give $G_M = J(1-\eta)(1+\varphi)/[(1-\eta)(1+\varphi)+\eta]$. By Proposition 2, $\partial P^*/\partial k = \rho \omega_D D(c)$ and $s_D' = -\lambda s_D(1-s_D)$, so $\varphi \to 0$ as $\omega_D \to 0$.

Proof of Proposition 3. (i) The Nash objective is single-peaked with unconstrained maximizer k^u, so its maximizer on $[0, \bar{k}]$ is $\min\{k^u, \bar{k}\}$. If $\bar{k} < k^u$, the objective is increasing on $[0, \bar{k}]$, so no $k < \bar{k}$ is agreed.

(ii) When the ceiling binds, $k = \bar{k}$ and the insurer chooses $P^*(\bar{k})$. By the envelope theorem, $d\Pi_I/d\bar{k} = \partial \Pi_I/\partial k = -Q$. The insurer's cost per user is $u_D = (\bar{k} - c)D(c)$, so by Proposition 2, $dP^*/d\bar{k} = \rho\, \omega_D D(c)$. Consumer surplus depends on $\bar{k}$ only through P, with $\partial CS_D/\partial P = -s_D$ and $\partial CS_N/\partial P = -s_N$, so $dCS_A/d\bar{k} = -S\rho\, \omega_D D(c)$. Manufacturer profit is $\Pi_M = \delta s_D(\bar{k} - m)D(c) + \delta(1-s_D)(p_L - m)D_0(p_L)$, so

$$\frac{d\Pi_M}{d\bar{k}} = Q + \delta s_D' \, \Delta_M \, \frac{dP^*}{d\bar{k}}, \qquad \Delta_M \equiv (\bar{k} - m)D(c) - (p_L - m)D_0(p_L).$$

Since $\delta s_D' = \omega_D S'$ by the definition of ω_D, the second term equals $\omega_D^2 \, S' \rho \, D(c) \, \Delta_M$, which is of order ω_D^2.

(iii) With $\omega_D = 0$, P^* does not vary with $\bar{k}$, so neither does any consumer surplus term or enrollment; $\Delta\Pi_I = -\Delta\Pi_M = Q$ per dollar, and total surplus is unchanged. With $\omega_D > 0$, each group's surplus rises by its enrollment times the premium reduction, $\delta s_D \rho\, \omega_D D(c)$ for users and $(1-\delta)s_N \rho\, \omega_D D(c)$ for non-users. Summing the changes in (ii), a \$1 reduction in $\bar{k}$ changes total surplus by

$$\rho\, \omega_D D(c)\big[S - \delta s_D' \Delta_M\big].$$

The insurer's first-order condition for P gives $S = -\delta s_D'\big[P - (\bar{k} - c)D(c)\big] - (1-\delta)s_N' P$, so the bracket equals $-\big[\delta s_D' \Omega_D + (1-\delta)s_N' P\big] = \Lambda > 0$, with $\Omega_D = P + (c-m)D(c) - (p_L - m)D_0(p_L)$ as in Proposition 6.

Proof of Proposition 4. With $D_0 \equiv 0$, $W_0 \equiv 0$ and a fixed copay, p_L does not appear in s_D, s_N, Π_I, Π_M, $\Pi_M^0 = 0$, or Π_I^0, all of which depend on (k, c, P). The insurer's premium problem given k is therefore independent of p_L, and so is the Nash objective as a function of

k. Its unconstrained maximizer $k^u(c)$ is independent of p_L, proving (i). For $p_L \geq k^u$, $k^* = \min\{k^u, p_L\} = k^u$, and every outcome is a function of $(k^*, c, P^*(k^*))$ only, proving (ii); the rebate rate is $1 - k^u/p_L$ by definition. For (iii), a list price $\bar{p}$ restricts $k \leq \bar{p}$ and, absent any payoff that depends on p_L, has no other effect; a net price ceiling $\bar{k}$ restricts $k \leq \bar{k}$. Both yield $k^* = \min\{k^u, \text{ceiling}\}$ by Proposition 3(i).

Proof of Proposition 5. The derivation uses the following notation: s_{md} and s_{0d} are insured and cash demand, $D(c)$ and $D_0(p_L)$ in the text; r is the rebate, so the net price is $k = p_L(1-r)$; α is the user share δ; and π_{mfg} is manufacturer profit with $m = 0$. The expression $r^* = 1 - s_{0d}/s_{md}$ is $k^* = p_L D_0(p_L)/D(c)$, and the derivatives below are those stated in the proposition. Assume the insurer has all bargaining power ($\eta = 1$) to derive closed form solutions. The manufacturer is held to its outside option, so its gains from agreement are zero.

This implies that the optimal rebate rate is

$$r^* = 1 - \frac{s_{0d}}{s_{md}}$$

which depends on the ratio of purchase probability for enrolled and non-enrolled diabetics.

The change in negotiated rebate given a change in the list price and copay are given by

$$\frac{dr^*}{dp_L} = -\frac{1}{s_{md}}\frac{\partial s_{0d}}{\partial p_L} > 0, \qquad \frac{dr^*}{dc} = \frac{s_{0d}}{s_{md}^2}\frac{\partial s_{md}}{\partial c} < 0.$$

An increase in list price increases the rebate proportional to the change in manufacturer profit due to decreases in demand among non-enrolled diabetic patients. An increase in copay decreases the rebate proportional to the change in insurer profit due to changes in insurance demand and insurer drug costs.

Manufacturer. Since the manufacturer is held to their disagreement payoff, we can simplify the manufacturer profit to

$$\pi_{mfg} = \alpha p_L s_{0d}$$

With a \$1 decrease in the list price p_L, the manufacturer's disagreement option changes, and the change in manufacturer profit is

$$\frac{\partial \pi_{mfg}}{\partial(-p_L)} = -\alpha\Big[s_{0d} + p_L\frac{\partial s_{0d}}{\partial p_L}\Big].$$

This is negative when $s_{0d} > -p_L\,\partial s_{0d}/\partial p_L$, which occurs when patients are not very sensitive to out-of-pocket prices, the list price is low, or uninsured demand is already high.

With a \$1 decrease in copay, the manufacturer profit does not change, since the rebate adjusts to hold the manufacturer to the disagreement payoff.

Insurer. Writing p_m for the premium P, s_m^{diab} and $s_m^{nondiab}$ for user and non-user enrollment s_D and s_N, and λ_m for the premium sensitivity λ: since the insurer optimally sets premiums, the insurer is only impacted through the direct and bargaining channels.

$$\pi_m = \max_{p_m} \alpha \cdot \underbrace{s_m^{diab} \cdot \Big[p_m - (p_L(1-r)-c)\cdot s_{md}\Big]}_{\text{Diabetic patients}} + (1-\alpha)\cdot \underbrace{s_m^{nondiab}\cdot p_m}_{\text{Non-Diabetic patients}}$$

Let $M \equiv p_m^* - (p_L(1-r)-c)s_{md}$ denote the insurer's margin per diabetic enrollee, and note that $\partial s_m^{diab}/\partial \Delta W = \lambda_m s_m^{diab}(1-s_m^{diab})$, $\partial \Delta W/\partial p_L = s_{0d}$, and $\partial \Delta W/\partial c = -s_{md}$. A \$1 decrease in the list price changes insurer profit by

$$\frac{d\pi_m}{d(-p_L)} = \underbrace{\alpha s_m^{diab}(1-r)s_{md}}_{\text{lower drug cost}} - \underbrace{\alpha\lambda_m s_m^{diab}(1-s_m^{diab})\,s_{0d}\,M}_{\text{lower enrollment}} - \underbrace{\alpha s_m^{diab} p_L s_{md}\cdot\frac{dr^*}{dp_L}}_{\text{lower rebate}}$$

and a \$1 decrease in the copay changes it by

$$\frac{d\pi_m}{d(-c)} = \underbrace{-\alpha s_m^{diab}\Big[s_{md} - (p_L(1-r)-c)\frac{\partial s_{md}}{\partial c}\Big]}_{\text{higher cost per fill and more fills}} + \underbrace{\alpha\lambda_m s_m^{diab}(1-s_m^{diab})\,s_{md}\,M}_{\text{higher enrollment}} - \underbrace{\alpha s_m^{diab} p_L s_{md}\cdot\frac{dr^*}{dc}}_{\text{higher rebate}}.$$

A net price ceiling $\bar{k}$ does not change p_L or $D_0(p_L)$ and so leaves the manufacturer's outside option unchanged; the equilibrium becomes $\min\{k^*, \bar{k}\}$.

Proof of Proposition 2. The insurer's first-order condition is $F(P; u_D, \tau) \equiv \delta[s_D + s_D'(P - u_D - \tau)] + (1-\delta)[s_N + s_N'(P-\tau)] = 0$. By the implicit function theorem, $\partial P^*/\partial u_D = -F_{u_D}/F_P = \delta s_D'/F_P$ and $\rho = \partial P^*/\partial \tau = -F_\tau/F_P = [\delta s_D' + (1-\delta)s_N']/F_P$. Dividing gives $\partial P^*/\partial u_D = \rho\,\omega_D$.

Under the logit, $s_g' = -\lambda s_g(1 - s_g)$, so for both groups insurance demand is decreasing in the premium, and the premium is a function of the marginal cost per unit of insured drug demand, scaled by the total enrollment sensitivity to premium changes.

Proof of Corollary 1. Holding k fixed and ignoring the valuation shift, a \$1 copay reduction raises u_D by $D(c) - (k - c)D'(c)$, so $dP^*/d(-c) = \rho\omega_D[D(c) + (k - c)|D'(c)|]$. Substituting into (16): users' surplus rises when $D(c) > dP^*/d(-c)$, which holds since $\rho\omega_D < 1$ and induced fills are small, and aggregate consumer surplus falls when the stated inequality holds. With $s_D = s_N$, $\omega_D = \delta$ and $D'(c) = 0$, the inequality becomes $\delta s D(c) < s\rho\delta D(c)$, that is, $\rho > 1$.

Proof of Proposition 6. Summing profits, $\Pi_I + \Pi_M = \delta s_D[P + (c - m)D(c)] + (1 - \delta)s_N P + \delta(1 - s_D)(p_L - m)D_0$, so k cancels and $TS = \delta CS_D + (1 - \delta)CS_N + \Pi_I + \Pi_M$ depends on k only through P. Using $\partial CS_D/\partial c = -s_D D(c)$, $\partial CS_D/\partial P = -s_D$, and $\partial CS_N/\partial P = -s_N$,

$$\frac{\partial TS}{\partial c}\Big|_P = \delta\big[-s_D D + s_D(D + (c - m)D')\big] + \delta\frac{\partial s_D}{\partial c}\big[P + (c - m)D - (p_L - m)D_0\big],$$
$$\frac{\partial TS}{\partial P} = \delta s_D'\,\Omega_D + (1 - \delta)s_N'\,P.$$

The total derivative is $\partial TS/\partial c|_P + (\partial TS/\partial P)(dP^*/dc)$.

Proof of Proposition 7. Under Assumption 1 there is no cash market, so $\Omega_D = P + (c - m)D(c)$, and $\partial s_D/\partial c = 0$, so the copay moves enrollment only through the premium. Write $Q = \delta s_D D(c)$, $\Pi_I = SP - (k - c)Q$ and $\Pi_M = (k - m)Q$. By the definition of ω_D, $\delta s_D' = \omega_D S' = -\omega_D|S'|$, and by Proposition 6, $\partial TS/\partial P = \delta s_D'\Omega_D + (1 - \delta)s_N' P = S'[P + \omega_D(c - m)D(c)] = -\Lambda$.

(a) A binding ceiling lowers k by \$1 with $D(c)$ fixed. By Proposition 3(ii), $\Delta\Pi_I = Q$, the premium falls by $\rho\,\omega_D D(c)$, and $\Delta CS_A = S\rho\,\omega_D D(c)$, of which users receive $\delta s_D\rho\,\omega_D D(c)$. Differentiating $\Pi_M = \delta s_D(k - m)D(c)$ along $dP^* = -\rho\,\omega_D D(c)$ gives $\Delta\Pi_M = -Q - (k - m)D(c)\,\delta s_D'\,\rho\,\omega_D D(c) = -Q + |S'|\rho\,\omega_D^2 D(c)^2(k - m)$, and $\Delta TS = -(\partial TS/\partial P)\,\rho\,\omega_D D(c) = \Lambda\rho\,\omega_D D(c)$.

(b) A \$1 copay reduction lowers k by κ and, by equation (15), raises the premium by ΔP in (23). By equation (16), users gain $Q - \delta s_D\Delta P$, non-users lose $(1 - \delta)s_N\Delta P$, and $\Delta CS_A = Q - S\Delta P$. Differentiating Π_M: the net price falls by κ on Q fills, fills rise by X, and user enrollment falls by $\delta s_D'\Delta P$, so $\Delta\Pi_M = -\kappa Q + (k - m)X - |S'|\,\omega_D\,\Delta P\,(k - m)D(c)$. By the envelope theorem for P, $\Delta\Pi_I = -(1 - \kappa)Q - (k - c)X$. Summing the three, and using the insurer's first-order condition $S = |S'|P - \omega_D|S'|(k - c)D(c)$, gives $\Delta TS = (c - m)X - \Lambda\Delta P$.

When $\omega_D = 0$, P^* responds to neither instrument, so $\Delta P = 0$, s_D and s_N are fixed, and Lemma 1 holds exactly: $k^u = m + (1 - \eta)J/Q$ with $J = \delta s_D[P + (c - m)D(c)] + (1 - \delta)s_N P - \Pi_I^0 + \varphi$.

Hence $J/Q = c - m + B/D(c)$ with $B = (SP - \Pi_I^0 + \varphi)/(\delta s_D)$ independent of c, and $dk^u/dc = (1-\eta)[1 - BD'(c)/D(c)^2] = (1-\eta)[1 + \epsilon_c B/(cD(c))] = \kappa$, which is equation (25); the payoff changes reduce to $\Delta CS_A = Q$, $\Delta\Pi_M = -\kappa Q + (k-m)X$, $\Delta\Pi_I = -(1-\kappa)Q - (k-c)X$ and $\Delta TS = (c-m)X$.

(c) With c fixed, a ceiling at θp_L lowers the net price from k^* to $\min\{k^*, \theta p_L\}$. Net spending is kQ, so on the initial fills it falls by $Q(k^* - \theta p_L)^+ = p_L Q(1 - r - \theta)^+$. By (a), the lower premium raises user enrollment by $-\delta s_D' \Delta P = \omega_D |S'| \Delta P$ with $\Delta P \propto \omega_D$, so induced fills, and the spending on them, are of order ω_D^2.

A copay cap at $\bar{c}$ moves the copay along $q \in [\bar{c}, c]$, with the premium at $P^*(q)$. By the envelope theorem, $\partial CS_A/\partial c = -\delta s_D D(c)$ and $\partial CS_A/\partial P = -S$, so along this path

$$\frac{dCS_A}{dq} = -\delta s_D(q) D(q) - S(q) \frac{dP^*}{dq}.$$

Integrating from $\bar{c}$ to c gives $CS_A(\bar{c}) - CS_A(c) = \int_{\bar{c}}^{c} \left[\delta s_D(q) D(q) + S(q)\, dP^*/dq\right] dq$, which is (24). When $\omega_D = 0$, $dP^*/dq = 0$ and s_D is constant, so (24) reduces to $\delta s_D \int_{\bar{c}}^{c} D(q)\, dq$, and adding and subtracting $D(c)$ inside the integral gives the decomposition into $Q(c - \bar{c})$ and the utilization response.

B Additional Results

B.1 Background Exhibits

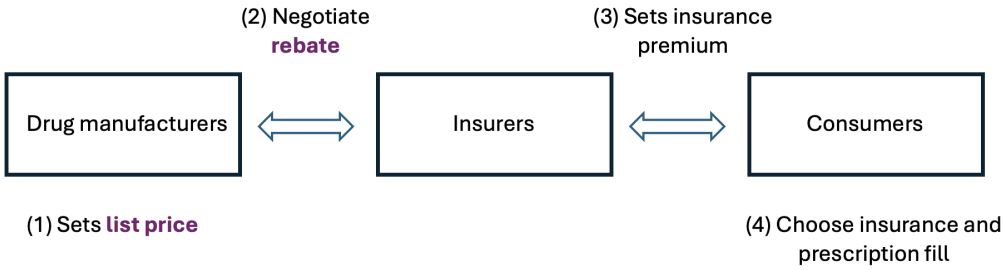


Figure A1: The Vertical Pricing Chain for a Branded Drug

B.2 Benchmark Model and What Changes Relative to It

Two features of the benchmark account for the inversion of the instrument ranking between the benchmark and the full model (Section 4.7). First, its upstream price works through a large cash margin. Under disagreement, consumers buy at the list price with probability 0.63, against 0.91 when covered, so $D_0/D(c) = 0.70$ and a lower list price forces the manufacturer to

Table A1: Instruments, the Prices They Regulate, and Their Scope

Policy	Price regulated	Scope	Model object
Insulin cost-sharing cap	Patient out-of-pocket price per 30-day supply	Covered insulin products in Part D and Part B; conditional on the plan covering the product	Copay c
Part D annual OOP cap	Patient liability in the tail of annual spending	All Part D drugs	Outside the static model
Maximum fair price	Price the manufacturer may charge for Medicare use, benchmarked to the prior net price	Selected single-source drugs without generic or biosimilar competition; Medicare only	Ceiling on k
Most-favored-nation agreements	Medicaid net prices, direct-to-consumer cash prices, launch prices	Participating manufacturers; channel-specific	Ceiling on k in one channel; cash price in another
State upper payment limits	Price payers may pay in the state	Selected drugs; Colorado's first limit is enjoined	Ceiling on k
Inflation rebates; Medicaid rebate-cap removal	List price, through a fiscal penalty on its level and growth	All payers; drove the 2023–24 insulin list-price cuts of 70%–78%	Environment: p_L

Notes: From 2026 the insulin copay cap is the lesser of \$35, 25% of the maximum fair price, and 25% of the plan's negotiated price, which links the first and third rows. The model's two instruments are the copay c and a net price ceiling, k; the list price p_L is a parameter of the environment that matters through cash purchases.

Table A2: Estimated Effects of the Part D Senior Savings Model ($35 Insulin Copay Cap)

	MA-PD		PDP	
Outcome	2021	2022	2021	2022
Insulin users				
30-day insulin fills per year	0.85	0.74	0.82	1.05
Annual OOP drug costs ($)	−214	−198	−364	−426
Annual OOP plus premium ($)	−209	−199	−309	−319
Enrollment in participating plans (%)	27.4	36.6	74.8	92.2
Non-users				
Annual OOP drug costs ($)	16	27	−18	−40
Annual OOP plus premium ($)	17	24	30	65
Manufacturers and Medicare ($ per member-month)				
Drug rebates	2.27	3.21	7.49	16.00
Coverage-gap discounts	1.84	2.76	4.06	7.96
Part D costs to Medicare	−4.74	−6.51	−9.33	−19.00

Notes: Difference-in-differences estimates for participating relative to comparison plans, from CMS (2025). MA-PD: Medicare Advantage prescription drug plan; PDP: stand-alone prescription drug plan. Non-user premium changes can be computed as the difference between the second and first non-user rows.

Table A3: Payoffs in the Benchmark Model

	Agree (drug covered at net price k)	Disagree
Manufacturer	$k\,D(c)$	$p\,D(p)$
Insurer	$\underbrace{[\alpha(W(c)-W(p))+(1-\alpha)(k-c)D(c)]}_{\text{Premium}}-(k-c)D(c)$	0
Consumers	$W(c)-\underbrace{[\alpha(W(c)-W(p))+(1-\alpha)(k-c)D(c)]}_{\text{Premium}}$	$W(p)$

Notes: The manufacturer's welfare is its revenue from drug sales. The insurer's welfare is its revenue from premiums minus its costs. Each consumer's welfare is their expected value from the drug minus their premium.

Table A4: What Each Instrument Can Deliver to Consumers

Policy at maximum stringency	Δ Consumer surplus	Δ Manufacturer profit	Δ Insurer profit	Consumer gain per $1 extracted
A. Benchmark, manufacturer makes the offer (per consumer)				
Upstream price down to the copay	+0.326	−0.326	0.000	1.00
Copay to zero	+0.000	+0.009	0.000	—
B. Benchmark, insurer makes the offer (per consumer)				
Upstream price down to the copay	+0.271	−0.216	−0.055	1.25
Copay to zero	+0.004	0	0.004	—
C. Full model, calibrated ($ per beneficiary-year)				
Upstream price to zero	+0.04	−11.38	+11.38	0.003
Copay to zero	+10.79	−5.15	−5.15	2.10

Notes: Panels A and B evaluate Proposition 1 at $D(x) = (1-x)^{1/2}$, $\alpha = 0.5$, $p = 0.6$, $c = 0.18$; the upstream price is the list (cash) price, which cannot fall below the copay without destroying the gains from coverage. Panel C is the calibrated model, with the upstream price the net price ceiling, driven to zero. The last column is R in equation (22); it is undefined where the manufacturer does not lose. Evaluated at marginal rather than maximal policies, R is 2.22 for a $1 copay cut and 0.003 for a ceiling at 75% of the bargained price.

concede. Our baseline sets this margin to zero and the cash extension to 3.6%, and a net price ceiling does not operate through it at all. Second, the benchmark's premium mechanically returns a share $1-\alpha$ of any surplus to consumers, so an insurer's saving is shared and a copay cut is clawed back. In the full model the premium is set by a profit-maximizing insurer and, by Proposition 2, responds to a drug-specific cost only in proportion to users' share of marginal enrollees. A copay cut is therefore a transfer that users keep, and a ceiling's saving stays with the insurer, unless users are marginal (Table 7).

B.3 Additional Results for the Application

Cash markets and the outside option. Figure A2 traces the bargained net price against a net price ceiling and against the list price. The ceiling leaves the outside option unchanged, so the net price is flat above the bargained price and falls one for one below it; a list-price cut behaves the same way only without a cash market. With inelastic cash demand ($\epsilon_0 = 0.5$) a lower list price weakens the outside option and the net price falls throughout; with elastic cash demand ($\epsilon_0 = 1.5$) the net price *rises* from $45.81 to $46.84 when the list price is cut to $500, as Proposition 5 predicts, and falls only once cash demand turns inelastic below about $450. With η near one the numerical solution reproduces the proposition's closed form ($23.78 versus $23.73).

Table A7 decomposes three policies into the channels of equation (8). For the copay cut, the direct channel costs the insurer $11.44 per beneficiary-year, the premium channel is negligible, and the bargaining channel returns $6.29 by lowering the net price. A ceiling at 75% of the

Table A5: Sensitivity Analysis

	Baseline			Copay from $30.90 to $0						
Specification	Net price	ω_D (%)	Δk at $200 list	Fills (%)	Premium ($/mo)	User ($/yr)	Non-user ($/yr)	Insurer ($/yr)	Mfr. ($/yr)	Total ($/yr)
Baseline	33.98	0.1	0.00	10.0	0.01	217	-0.08	-5.15	-5.15	0.49
Aggregate OOP elasticity 0.05	34.02	0.0	0.00	5.0	0.00	213	0.00	-5.19	-5.19	0.26
Aggregate OOP elasticity 0.16	33.60	0.5	0.00	17.3	0.10	219	-0.67	-4.84	-4.84	0.63
Premium pass-through 0.13	21.10	0.0	0.00	9.8	0.00	218	0.00	-5.19	-5.19	0.51
Premium pass-through 0.74	42.95	1.2	0.00	12.4	0.33	206	-1.16	-4.61	-4.61	-0.02
Insurer bargaining weight 0.2	54.35	0.1	0.00	10.0	0.01	217	-0.06	-2.06	-8.24	0.51
Insurer bargaining weight 0.8	13.59	0.1	0.00	10.0	0.02	217	-0.10	-8.24	-2.06	0.47
Insulin-user share 0.02	34.37	0.0	0.00	10.1	0.01	217	-0.04	-2.06	-2.06	0.20
Insulin-user share 0.10	33.28	0.1	0.00	10.0	0.02	217	-0.11	-10.33	-10.33	0.99
Coverage value $25/month	28.85	3.4	0.00	24.3	0.65	179	-4.21	-2.55	-2.54	-0.16
Marginal cost $5 per fill	36.48	0.1	0.00	10.0	0.01	217	-0.08	-5.23	-5.23	0.32
Cash market, elasticity 0.5	45.69	0.2	-7.02	10.6	0.04	216	-0.28	-5.06	-5.06	0.40
Cash market, elasticity 1.5	45.81	0.1	-1.99	10.1	0.02	217	-0.12	-5.14	-5.14	0.47
Net price benchmarked to a 70% rebate	199.35	0.1	0.00	10.0	0.00	218	0.02	-5.15	-5.15	0.60

Notes: Each row re-calibrates (λ, g) to a $48 premium and the stated pass-through (0.45 unless noted). Δk at $200 list is the change in the bargained net price when the list price is cut to $200. Insurer, manufacturer and total are per beneficiary-year.

Table A6: Counterfactual Copays

	Net			Fills	Δ Consumer surplus			Δ per beneficiary-year		
Copay	price ($)	Rebate (%)	Premium ($/mo)	per user (/mo)	User ($/yr)	Non-user ($/yr)	All ($/yr)	Insurer ($)	Mfr. ($)	Total ($)
$30.90 (baseline)	33.98	94.9	48.00	0.558	0.0	0.00	0.00	0.00	0.00	0.00
$25	30.69	95.4	48.00	0.569	39.9	-0.02	1.97	-0.90	-0.90	0.17
$20	27.92	95.8	48.01	0.578	74.3	-0.04	3.68	-1.69	-1.69	0.29
$10	22.40	96.6	48.01	0.596	144.7	-0.06	7.18	-3.37	-3.37	0.44
$0	16.91	97.5	48.01	0.614	217.3	-0.08	10.79	-5.15	-5.15	0.49

Notes: List price fixed at $664.50. Net price and premium are equilibrium outcomes. Fills per user include enrollment. Changes are relative to the baseline copay of $30.90.

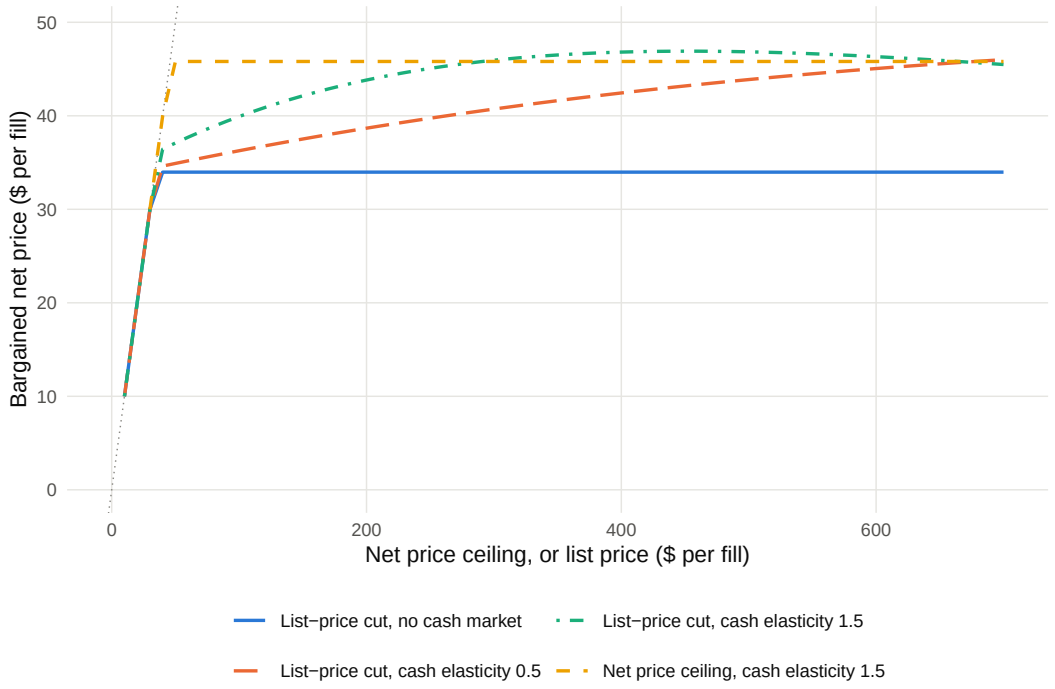


Figure A2: Bargained Net Price under List-Price Cuts and Net Price Ceilings

Notes: Copay fixed at $30.90. The dotted diagonal is the 45-degree line. Cash-market specifications are recalibrated to the same premium and pass-through targets. Without a cash market the net price is flat in the list price until the list price falls below the bargained net price (Proposition 4). With a cash market the list price enters the manufacturer's outside option and the value of going uninsured, so the net price varies with it everywhere: it rises throughout when cash demand is inelastic, and peaks near $450–$500 when cash demand is elastic at the list price, where cash revenue is highest. A net price ceiling leaves the cash price unchanged and so is flat above the bargained price (Proposition 3).

Table A7: Decomposition into Direct, Premium, and Bargaining Channels

	Direct	Premium	Bargaining	Total
A. Copay cut from $30.90 to $0				
Net price ($/fill)	0.00	0.00	-17.07	-17.07
Premium ($/month)	0.00	0.01	0.00	0.01
Insulin-user surplus ($/user-year)	217.48	-0.16	0.01	217.33
Non-user surplus ($/non-user-year)	0.00	-0.09	0.01	-0.08
Insurer profit	-11.44	0.00	6.29	-5.15
Manufacturer profit	1.14	0.00	-6.29	-5.15
Total surplus	0.58	-0.09	0.01	0.49
B. Net price ceiling at 75% of the bargained net price				
Net price ($/fill)	-8.49	0.00	0.00	-8.49
Premium ($/month)	0.00	0.00	0.00	0.00
Insulin-user surplus ($/user-year)	0.00	0.02	0.00	0.02
Non-user surplus ($/non-user-year)	0.00	0.01	0.00	0.01
Insurer profit	2.85	0.00	0.00	2.85
Manufacturer profit	-2.85	0.00	0.00	-2.85
Total surplus	0.00	0.01	0.00	0.01
C. List-price cut to $200, no cash market				
Net price ($/fill)	-23.75	0.00	23.75	0.00
Premium ($/month)	0.00	0.00	0.00	0.00
Insulin-user surplus ($/user-year)	0.00	0.05	-0.05	0.00
Non-user surplus ($/non-user-year)	0.00	0.03	-0.03	0.00
Insurer profit	7.96	0.00	-7.96	0.00
Manufacturer profit	-7.96	0.00	7.96	0.00
Total surplus	0.00	0.03	-0.03	0.00

Notes: Direct: new policy, with the baseline rebate rate and premium (for Panel B, the net price set at the ceiling). Premium: premium re-optimized at the baseline rebate rate. Bargaining: rebate renegotiated. Insurer, manufacturer and total surplus are per beneficiary-year.

bargained price operates entirely through the direct channel. A list-price cut to $200 without a cash market shows neutrality channel by channel: holding the rebate *rate* fixed would transfer $7.96 per beneficiary-year to the insurer, and renegotiation undoes all of it.

B.4 Insurance Demand Estimates

This appendix reports estimates of plan choice, which can discipline the premium sensitivity λ and the value of insulin coverage in Section 6 directly. Because the estimates are unstable across specifications, with the coefficients on insulin coverage and on premiums varying in sign, the paper calibrates these parameters to the premium and to external pass-through estimates instead; we report the baseline specification.

Sample. We filter to non-subsidized individuals enrolled in stand-alone Part D plans or not in any drug prescription plans in January 2022. We exclude individuals enrolled in Medicare Advantage plans.

The ex-ante value of insurance j's coverage of insulin for patients in demographic group g using length of action l as

$$W_{gjl} = \ln\left(1 + \sum_{j \in J_l} \exp(\delta_{jgt})\right)$$

where $l = 0$ indicates someone who is not diabetic (no insulin usage in a given year). We then create a singular expected value of insulin coverage for each plan by weighing W_{glt} using the annual share of individuals in each LOA group.

Patients choose insurance plan depending on the plan's premiums and drug coverage (with county and insurer FEs):

$$\begin{aligned} v_{jlt} &= \underbrace{\beta_g W_{jl}}_{\text{value from insulin coverage}} + \underbrace{\beta_{jl}}_{\text{other ins value}} - \underbrace{\phi_{jt}}_{\text{premium disutility}} + \eta_{jglt} \\ &= \delta_{jglt} + \eta_{jglt} \end{aligned}$$

The outside option is no insurance, and we assume patients without insurance pay the list price:

$$v_{0glt} = \beta_g W_{0gl} + \eta_{0glt}$$

The group and length of action specific market share for insurance m is thus

$$s_{jglt} = \frac{\exp\left(\delta_{jglt}\right)}{\sum_{j' \in \{0,J\}} \exp\left(\delta_{j'glt}\right)}$$

We exclude EWGP and PACE plans. To construct patient out-of-pocket cost for each drug, we used the pre-ICL 3 month estimates. For plans that did not report a pre-ICL 3 month copay or coinsurance, we assumed a standard 25% coinsurance.[6]

Table A8 shows the baseline estimates for insurance demand. Plan characteristics include deductible, formulary, and reduced cost sharing. Specifications with demographic heterogeneity, and specifications instrumenting premiums with the average ΔW and deductible of other plans in the same region, yielded coefficients of varying sign on insulin coverage and on premiums, which is why we do not use them.

Table A8: Insurance Demand Estimates (No Heterogeneity)

	(1)
Insulin coverage (ΔW_{jgl})	0.020*
	(0.009)
Premium (00s)(ϕ_{jt})	−0.021***
	(0.004)
Implied WTP for Mean Plan (Quarter)	75
Observations	2,298
Plan Characteristics	Y
R^2	0.26

C Additional Data Description

This appendix collects the remaining descriptive exhibits for the 2022 Part D sample of Section 6: beneficiary characteristics, the medication classes used to identify likely diabetics, insulin fill types and plan characteristics, and the distributions of insulin coverage, cost sharing, and formulary placement across stand-alone plans.

[6] This was imputed for 93,073 (out of 149,648) drug-insurance pairs.

Table A9: Beneficiaries Summary Statistics

Characteristic	All	Part D Only
Share	1	0.79
Age	72	72
Female	0.54	0.57
Race/Ethnicity		
White	0.78	0.78
Black	0.11	0.11
Hispanic	0.03	0.04
Other	0.08	0.08
Share with any fill	0.15	0.19
Share with potential insulin fill	0.04	0.05
Share with insulin fill	0.01	0.01

Table A10: Medication Classes Used to Identify Diabetes-Related Pharmacotherapy in Medicare Part D Claims (2022)

Primary Clinical Use	**Included Search Words**
Insulin therapy (type 1 and type 2 diabetes)	insulin, aspart, lispro, glulisine, glargine, detemir, degludec, isophane
Oral and non-insulin injectable glucose-lowering agents (type 2 diabetes)	glipizide, glimepiride, glyburide, sitagliptin, linagliptin, saxagliptin, alogliptin, janumet, tradjenta, repaglinide, nateglinide, pioglitazone, rosiglitazone, acarbose, miglitol, pramlintide
GLP-1 receptor agonists and dual incretin agents (type 2 diabetes; weight management)	lixisenatide, liraglutide, semaglutide, dulaglutide, exenatide, rybelsus, tirzepatide, mounjaro
Biguanide (type 2 diabetes; prediabetes; insulin resistance)	metformin

Table A11: Insulin Fill Summary Statistics (Type)

Characteristic	Share of Patient-Months
Observed insulin fill	0.56
Type of insulin	
Pen only	0.43
Vial only	0.09
Both	0.03
Length of action	
Rapid only	0.09
Short only	0.01
Intermediate only	0.03
Long only	0.26
Multiple	0.10

Table A12: Part D Plans Summary Statistics

Characteristic	Average	25th	75th
Standalone Part D Plans per Region	20	1	27
Premium ($, Monthly)	48	6	207
Deductible	369	0	480

Figure A3: Average Drug Coverage by PDP

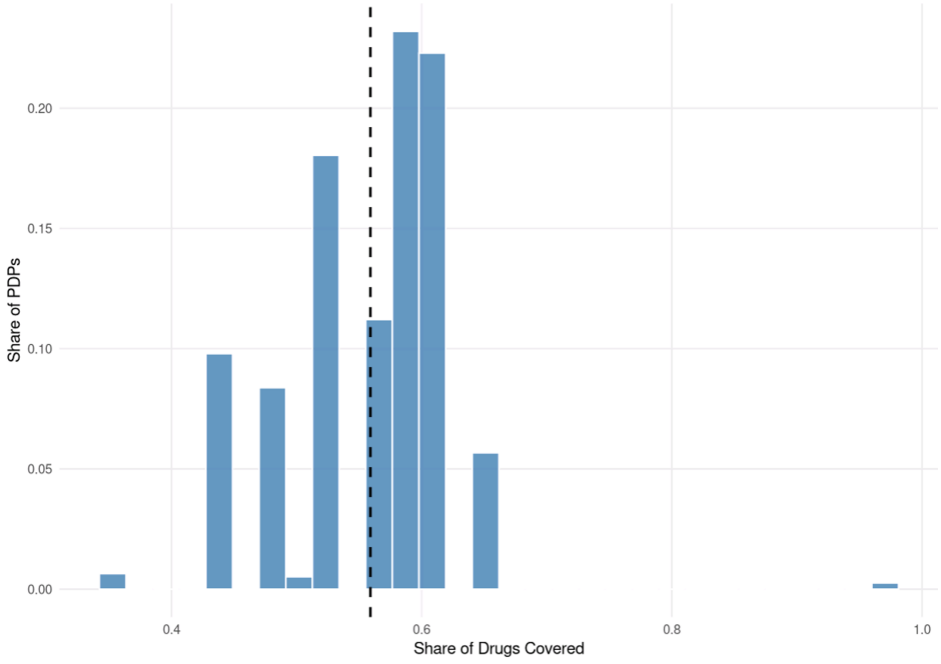
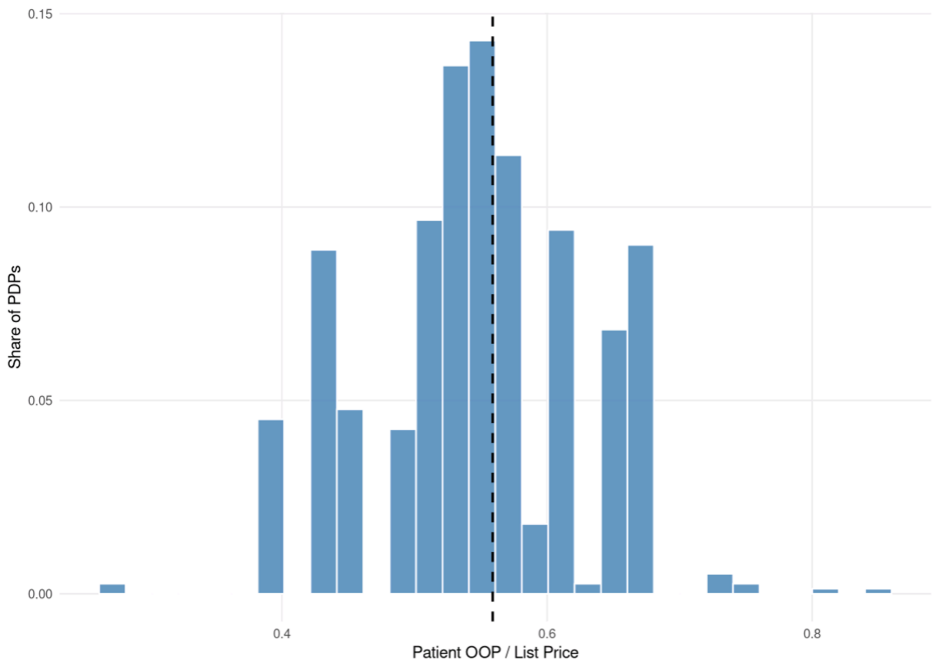


Figure A4: Average Patient Cost Sharing by PDP



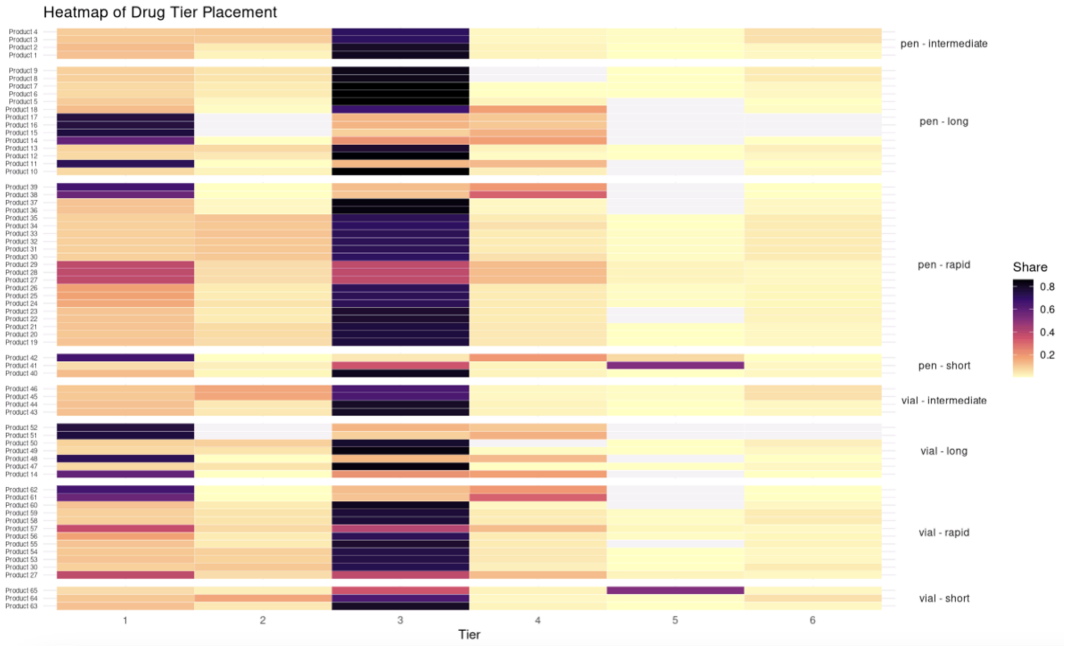


Figure A5: Distribution of Formulary Placement

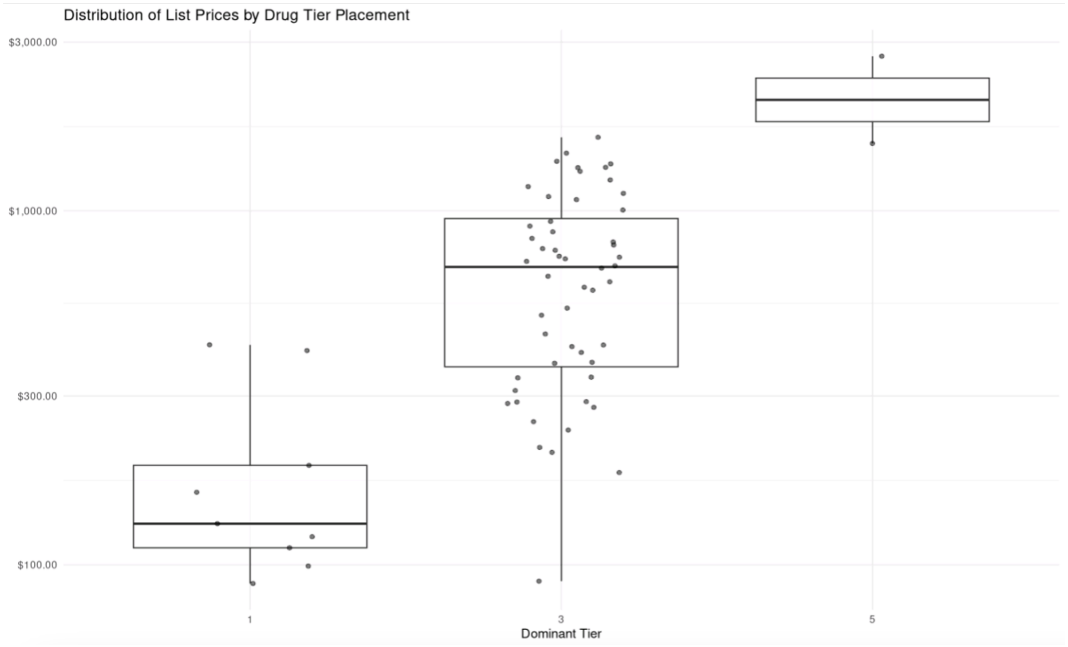


Figure A6: Distribution of List Prices by Dominant Drug Formulary Tier

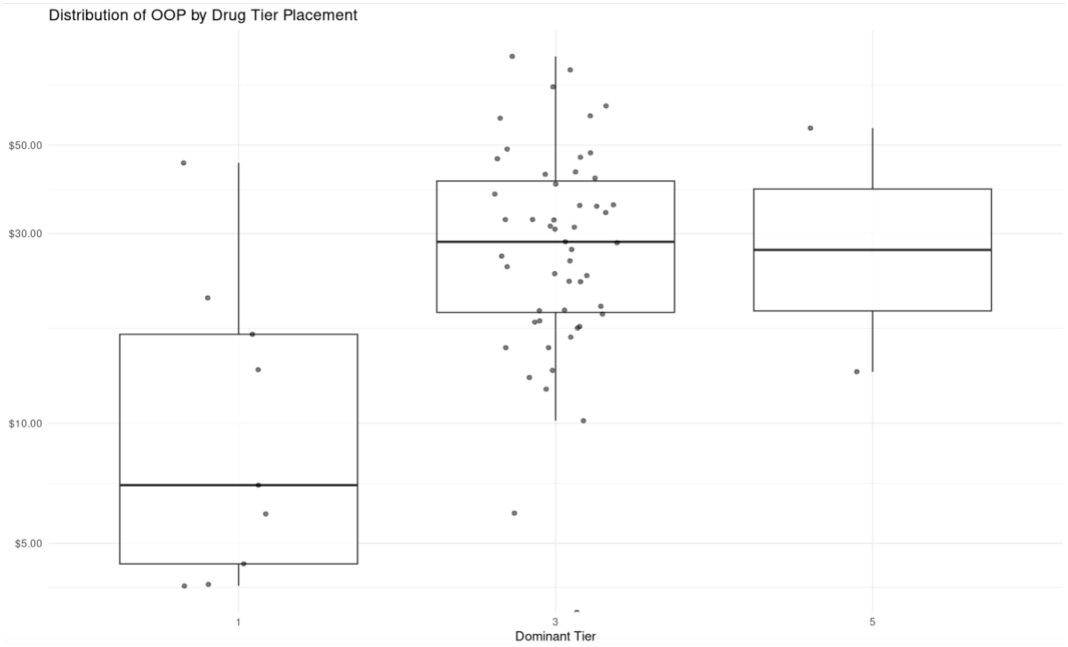


Figure A7: Distribution of Out-of-Pocket Prices by Dominant Drug Formulary Tier